

Task assignment as dynamic incentives^{*}

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Abstract

We study repeated task assignment as an instrument for providing effort incentives. Unlike traditional incentive instruments, assignment of a task both determines who produces and provides incentives, and incentives for one worker spill over to others because assignment is exclusive. We show that optimal incentives uniquely require a strict and evolving priority ranking through which workers are assigned the task. This ranking implies that workers' average workloads differ even when they are symmetric in all payoff-relevant respects. We characterize how workforce size, monitoring, and working conditions shape the scope of optimal incentive provision and the resulting inequality among workers.

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1 Introduction

Organizations routinely assign tasks to workers: sales managers allocate territories, consulting firms staff projects, hospitals assign residents to shifts, and gig-economy platforms dispatch routes. Prior work largely views task assignment as exploiting worker heterogeneity; see, e.g., [Garicano and Van Zandt’s \(2013\)](#) survey. This perspective abstracts from a key feature of many organizations: task assignment is a powerful incentive instrument, especially when monetary rewards are constrained. Desirable assignments such as premium sales territories, flagship projects, or high-visibility clients, and the relief from undesirable assignments such as crisis engagements, turnaround sales accounts, or low-value clients, reward strong performance.

In this paper, we characterize optimal incentive provision through task assignment and draw implications for organizational design. Our characterization departs sharply from standard incentive theory. Optimal assignment disregards relative worker comparison, in contrast to [Board \(2011\)](#) and [Andrews and Barron \(2016\)](#). Moreover, it induces unequal workloads across workers even when they are technologically independent and symmetric in all payoff-relevant respects, highlighting that the effort-equality tension is more pervasive than previously established ([Winter, 2004](#); [Halac, Lipnowski and Rappoport, 2021](#)).

We study a long-run allocation relationship without transfers. A manager repeatedly interacts with a set of ex ante identical workers. We begin with a baseline model where the manager has a single task that is undesirable for the workers. In each period, she assigns one worker to perform the task. This assignee privately chooses whether to exert effort or to shirk, producing a public noisy output. Workers prefer to be unassigned since the task is undesirable. When assigned, workers strictly prefer to shirk; the manager prefers effort. The manager has no commitment power; her strategy, which we call an assignment rule, is flexible and can depend on all past outputs and assignments in arbitrary ways. We focus on (the manager’s) first-best equilibria where workers exert effort when assigned on path.

Task assignment functions differently from canonical incentive instruments. Unlike contests or auctions that allocate fixed prizes, assignment both allocates the task and motivates effort. Unlike monetary contracts and information design, assignment is rivalrous: assigning one worker excludes others, so incentives for one necessarily spill over to others. Each worker’s effort incentives when assigned are intrinsically dynamic:

in any first-best equilibrium, these incentives stem from his future workload, namely his discounted frequency of being assigned. Because the task is undesirable, these incentives require that he receives a sufficiently lighter future workload following a current good output than a bad one, with future assignments structured in a way that preserves all workers’ effort incentives.

We characterize all uniformly first-best assignment rules for any given number of workers; these rules sustain a first-best equilibrium simultaneously for all parameters—output noise, payoffs, and discounting—for which a first-best equilibrium exists. This uniformity criterion entails no loss of generality for characterizing first-best equilibria and is theoretically attractive: by definition, a uniformly first-best rule sustains a first-best equilibrium whenever such an equilibrium exists. It is also practically attractive: it precludes fine-tuned assignment in the spirit of the Wilson doctrine (1987), reflecting that managers in practice rarely know the precise details of agency relationships.

Our main result shows that uniformly first-best assignment rules exist and all of them take the form of “first-order rotation:” each worker, when assigned, is retained after a bad output and is relieved after a good output until all others have been assigned and produced a good output. These rules are unique up to how workers are initially ordered and provide identical incentives. While it is intuitive that first-order rotation sustains first-best equilibria when effort cost is low, our result establishes a much stronger conclusion: these rules dominate all other rules for sustaining first-best equilibria whenever such an equilibrium exists, despite the manager’s flexibility. Moreover, remarkably, these rules disregard relative worker comparison despite the rivalrous nature of task assignment.

This result offers a perspective on the widespread practice of rotating workers through unpleasant tasks on the basis of first-order performance. For instance, hospitals commonly rotate residents through intensive shifts, with relief following successful completion and continued responsibility when complications arise. Sales representatives tend to remain responsible for unresolved accounts until successful resolution. While these practices are typically driven by operational needs, such as continuity of care or logistical considerations, our result establishes the incentive benefits that such structures can generate.

Our result implies that the scope for attaining first-best equilibria is fully determined by the effectiveness of first-order rotation in sustaining them. This tractability yields sharp comparative statics. Importantly, we show that improving effort moni-

toring may perversely reduce the scope for attaining first-best equilibria; in contrast, reducing the gain from shirking, improving the monitoring of shirking, increasing workforce size, and widening the payoff gap between assignment and unassignment all unambiguously expand this scope. Practically, these findings suggest that organizations can enhance incentive provision by maintaining a sizable workforce, ensuring unpleasant tasks to be engagement-heavy so that shirking provides limited comfort, incorporating progress-tracking mechanisms to enhance shirking detection, and offering unassigned workers genuine relief; at the same time, instruments that improve effort detection, such as granting assignees greater discretion that raises the productivity of effort, should be adopted with caution.

Our result highlights a tension between effort incentives and worker equality. We show that in any first-best equilibrium corresponding to first-order rotation, in all but possibly a finite number of initial periods on path, workers' payoffs, which reflect their workloads, are strictly ranked even when they are symmetric in all payoff-relevant respects—ex ante identical, playing identical strategies, and having identical output records. This is because in initial periods workers who have not yet been assigned can be assigned uniformly at random, but eventually all workers are assigned according to a strict rotation order. To gain further insights, we show that for parameters given which moral hazard is sufficiently mild, first-best equilibria where symmetric workers receive equal workloads exist, in addition to those corresponding to first-order rotation. Taken together, our results show that to sustain first-best equilibria, discriminatory workloads can be optimal when the moral hazard problem is sufficiently mild, and must be so otherwise. In practice, nonmonetary discrimination is pervasive in organizations.¹ Discriminatory workloads, in particular, are often attributed to worker characteristics (see, e.g., [Babcock, Recalde, Vesterlund and Weingart, 2017](#)); our results provide a perspective based purely on dynamic incentive provision.

We then study a variant of the model where the task is desirable for the workers. We show that our main result and its subsequent implications, namely the discriminatory workloads and the comparative statics, extend except that the quality of monitoring of effort has fundamentally different implications for effort incentives. Specifically, we show that first-order rotation continues to be the unique class of uniformly first-best assignment rules, with a reverse logic of rotation: an assignee is relieved after a

¹See, e.g., Adi Gaskell, “How non-financial rewards widen workplace inequality,” *Forbes*, July 7, 2021.

bad output but retained after a good one, since assignment here is a reward rather than a punishment. Similar to the undesirable-task setting, this sheds light on the common practice of allocating valuable tasks based on first-order performance: in consulting and sales, for example, strong performers continue to remain on prominent projects or accounts, while weaker performers are reassigned. However, unlike with an undesirable task, improving the monitoring of effort on a desirable task unambiguously expands the scope for attaining first-best equilibria. This distinction shows that the incentive implications of better monitoring of effort on an assignment depend on whether workers view the assignment as a reward or a burden, offering a perspective on why organizations often invest heavily in performance measurement for prestigious roles while relying on staffing policies and shirking detection for unpleasant tasks.

Finally, we study several other variants of our model that fit different applications. We show that our results extend, combining the logic in the undesirable-task baseline model and the desirable-task variant, to a setting in which the manager assigns different tasks, including both desirable and undesirable ones, over time. We also show that our results extend if workers face private payoff shocks, or if they have limited information—they may observe neither the assignee’s identity nor output unless they themselves are assigned, and may not know the workforce size.

Methodologically, our analysis overcomes a known difficulty in repeated principal-multi-agent games without transfers. Our results involve characterizing, for any number of workers, the largest set of parameters for which a first-best equilibrium exists. As noted in [De Clippel, Eliaz, Fershtman and Rozen \(2021\)](#), such characterization has not been obtained in the literature because the geometry of the equilibrium payoff set is not well understood, making it difficult to identify its extreme points that determine the scope for sustaining first-best equilibria. We circumvent this difficulty by simultaneously solving for the boundary of the largest set of parameters for which a first-best equilibrium exists and the extreme first-best equilibrium payoffs when parameters are on this boundary.

Related literature. We contribute primarily to the literature on dynamic multi-agent assignment. Our model is most closely related to [Board \(2011\)](#) and [Andrews and Barron \(2016\)](#) who also study assignment under dynamic moral hazard. Their dynamics are driven by observable, time-varying worker heterogeneity and the availability of transfers, both of which are absent in our model. Their optimal dynamics differ from

first-order rotation in two fundamental respects. First, they feature a form of favoritism where the manager tends to retain a worker until a significant shock regarding workers' heterogeneity occurs. Second, they rely on relative worker comparison: in Board (2011), assignment tends to favor workers who have previously produced for the manager and among them those with lower effort costs; in Andrews and Barron (2016), assignment tends to favor workers who have the highest productivity and among them those who have produced a good output most recently. De Clippel et al. (2021) study assignment among ex ante identical workers without transfers, but under adverse selection rather than moral hazard. They focus on assignment rules that treat currently unassigned workers symmetrically in each period; in our model, symmetric treatment of unassigned workers undermines effort incentives.

The effort-equality tension under moral hazard was first articulated in public static monetary contracts where a manager with commitment power uniquely implements effort among complementary workers (Winter, 2004; Halac et al., 2021), a setting that differs substantially from ours. In our model, the effort-equality tension is self-enforcing, dynamic, nonmonetary, arises among independent workers, and is robust to workers' uncertainty about others' assignments and outputs. A key takeaway from our results is therefore that this tension is more pervasive than previously recognized.

Our analysis contrasts with prior work on dynamic single-agent assignment (Li, Matouschek and Powell, 2017; Bird and Frug, 2019; Guo and Hörner, 2020; Lipnowski and Ramos, 2020). In these models, assignment is not rivalrous: the principal decides whether or not to assign, but not whom to assign. Our analysis also contrasts with models of dynamic moral hazard in teams (Che and Yoo, 2001; Rayo, 2007; Bonatti and Hörner, 2011; Georgiadis, 2015). In these models, workers are complementary and there are no assignment dynamics.

In practice, assignments that largely disregard past performances are common alongside performance-contingent ones. In our model, such assignments are feasible but do not address moral hazard. A related literature has shown that non-performance-contingent rotation schemes can be valuable when workers have hidden information (see, e.g., Ickes and Samuelson, 1987; Meyer, 1994; Ortega, 2001; Arya and Mittendorf, 2004; Arya and Mittendorf, 2006a,b; Prescott and Townsend, 2006; Eliaz, Fershtman and Frug, 2025). The literature has also studied the role of such schemes in limiting the formation of collusive relationships in political economy (Iyer and Mani, 2012) and in accounting (Lennox, Wu and Zhang, 2014).

2 Baseline model

We begin with a baseline model before extending it in various directions. Time $t = 0, 1, \dots$ is discrete and the horizon is infinite. There is a manager and a set of ex ante identical workers $N := \{1, \dots, n\}$, where $n \geq 1$. The manager is endowed with a task that is undesirable for the workers.

In each period the manager publicly assigns one worker to perform the task.² The assigned worker privately chooses whether to exert effort or to shirk. Effort yields a good output $\bar{y}$ with probability $p \in (0, 1)$ and a bad output $\underline{y}$ otherwise; shirking yields a good output with probability $q \in (0, p)$ and a bad output otherwise. This output is publicly observable.

In each period t , each worker i 's payoff, u_t^i , and the manager's payoff, v_t , are as follows. The assigned worker gets payoff normalized to zero if he exerts effort and payoff $s > 0$ if he shirks. Therefore, this worker prefers to shirk than to exert effort. Each unassigned worker receives a "resting" payoff $r > 0$. The manager's payoff depends only on the output; it is equal to $\bar{v}$ if the output is good and is equal to $\underline{v}$ if the output is bad, with $\underline{v} < \bar{v}$ so that she prefers a good output to a bad one.

A public history in each period t , h_t , is an element in $(N \times \{\bar{y}, \underline{y}\})^t$, consisting of the identities of previous assignees and their outputs, with $h_0 := \emptyset$. Worker i 's strategy is a collection $\sigma^i \equiv (\sigma_t^i)_{t=0}^\infty$, where $\sigma_t^i(h_t) \in [0, 1]$ specifies the probability that worker i exerts effort when assigned at history h_t . Thus, workers do not condition their strategies on past private actions; we explain at the end of this section that this restriction to public strategies is innocuous for our results. The manager's strategy is a collection $f \equiv (f_t)_{t=0}^\infty$, where $f_t(h_t) \in \Delta(N)$ specifies a distribution over workers from which the current assignee is drawn at history h_t . Therefore, her assignment is flexible and may depend on all past assignments and outputs in arbitrary ways.

The workers and the manager share a common discount factor $\delta \in (0, 1)$.³ Each worker i 's and the manager's average realized payoffs are

$$(1 - \delta) \sum_{t=0}^{\infty} \delta^t u_t^i \quad \text{and} \quad (1 - \delta) \sum_{t=0}^{\infty} \delta^t v_t.$$

²This rules out the possibility that the manager assigns no worker. This assumption is innocuous for our results. We shall focus on the manager's first-best equilibria where the manager would be worse off by assigning no one in any period on path.

³Our results go through if the manager has a different discount factor.

We use perfect public equilibrium (Fudenberg, Levine and Maskin, 1994), henceforth equilibrium, as the solution concept. An equilibrium exists: repeated play of static Nash equilibrium, such as one in which workers shirk when assigned and the manager assigns worker 1, is an equilibrium.

An equilibrium is first-best for the manager, or simply first-best, if each worker exerts effort when assigned on path. We call the manager’s strategy an assignment rule, and refer to the manager’s strategy in any first-best equilibrium as a first-best assignment rule. We do not impose any restriction on parameters to ensure that effort is efficient, namely that in each period the sum of the manager’s and the workers’ payoffs conditional on the assignee exerting effort is at least its counterpart conditional on the assignee shirking. If effort is efficient, then an equilibrium is efficient if assignees exert effort on path, and first-best equilibria and efficient equilibria are equivalent.

Our interest is in first-best assignment rules that are practically appealing by virtue of not being parameter-sensitive in the spirit of the Wilson doctrine (1987), as motivated in the Introduction. We formalize this idea through a robustness criterion in Definition 1 below. For any number n of workers, let

$$\Omega(n) := \{(p, q, r, s, \delta) \in (0, 1)^2 \times \mathbf{R}_{++}^2 \times (0, 1) : p > q\} \quad (1)$$

be the set of feasible parameter vectors, with typical element ω , and let $\Omega^*(n)$ be the largest set of such parameter vectors for which a first-best equilibrium exists. We call each $\omega \in \Omega^*(n)$ a first-best parameter vector.

We study first-best equilibria where the manager uses a uniformly first-best assignment rule:

Definition 1 (Uniform first-best). *For each n , an assignment rule is said to be uniformly first-best if it is first-best simultaneously for all $\omega \in \Omega^*(n)$.*

Therefore, given any uniformly first-best assignment rule, whenever a first-best equilibrium exists, there is a first-best equilibrium in which the manager plays this uniformly first-best rule. Consequently, the uniformity criterion is not only practically appealing but also theoretically so: no assignment rule outperforms uniformly first-best rules for attaining first-best equilibria.⁴ Our main result establishes the existence of

⁴The uniformity criterion is stronger than the Blackwell criterion (Blackwell, 1962; Cavounidis, Ghosh, Hörner, Solan and Takahashi, 2025). The Blackwell criterion would require strategies to attain the manager’s first-best in equilibrium for sufficiently high discount factors simultaneously. Uniformly first-best assignment rules are Blackwell first-best, but the converse is false.

uniformly first-best rules and characterizes all of them.

Since workers' actions are hidden, only the manager may have observable deviation in equilibrium. Since the manager attains her first-best (expected) payoff $p\bar{v} + (1-p)\underline{v}$ in each period and so has no profitable deviation on path in any first-best equilibrium, the specification of off-path behavior does not affect equilibrium existence or incentives. For concreteness, we assume that off path, workers shirk when assigned and the manager assigns worker 1. Hereafter, we omit mentioning off-path behavior for conciseness and focus on workers' incentives. This discussion also implies that granting the manager commitment power to an assignment rule does not change our results.

Finally, we explain why assuming that workers play public strategies is innocuous for our results. [Fudenberg and Levine \(1994\)](#) show that in repeated games with imperfect public product-structure monitoring, such as ours, the set of sequential equilibrium payoffs coincides with the set of perfect public equilibrium payoffs even when players' strategies may depend on their past private actions. Thus, insofar as we are concerned with equilibrium payoffs, there is no loss of generality in focusing on public strategies. There is also no loss of generality for equilibrium behavior: in any first-best equilibrium, assignees exert effort on path, regardless of whether strategies are restricted to be public.

3 Effort incentives

In this section, we present a preliminary result formalizing workers' incentives. In any equilibrium, denote worker i 's continuation payoff at history h_t by

$$U^i(h_t) := \mathbf{E} \left[(1 - \delta) \sum_{\tau=t}^{\infty} \delta^{\tau-t} u_{\tau}^i \middle| h_t \right],$$

where the expectation is taken with respect to the equilibrium outcome distribution. Let $h_t i y$ be the public history that is a concatenation of history h_t followed by an assignment of worker i who then produces output y .

Lemma 1 (Incentives). *In any first-best equilibrium, at any history h_t on path where worker i is assigned with positive probability,*

$$U^i(h_t i \bar{y}) - U^i(h_t i \underline{y}) \geq \frac{(1 - \delta)s}{\delta(p - q)}, \quad (2)$$

and

$$\sum_{j=1}^n U^j(h_t) = (n-1)r. \quad (3)$$

Appendix A collects the proofs of all formal results. The incentive constraint (2) formalizes that effort incentives are intrinsically dynamic. If the interaction were one-shot, each worker would have a strict incentive to shirk when assigned; therefore, a worker is willing to exert effort when assigned only if his continuation payoff upon a good output is sufficiently higher than that upon a bad output, namely (2) holds. The rivalry constraint (3) highlights that raising one worker’s continuation payoff necessarily reduces at least one other worker’s continuation payoff. This constraint holds because in any first-best equilibrium, in each period on path, the assignee obtains payoff 0 while the other $n-1$ workers obtain payoff r .

Lemma 1 highlights that assignment dynamics in any first-best equilibrium are driven solely by the incentive constraint and the rivalry constraint at each history. In each on-path continuation of any first-best equilibrium, a worker’s payoff reflects his workload, namely his average discounted frequency of being assigned. The incentive constraint requires that in each such continuation, future assignments are structured so that current assignees face a sufficiently lighter workload following a good output than following a bad output. The rivalry constraint further requires that these future assignments are structured in a way that preserves all future assignees’ effort incentives on path. Consequently, as described in the Introduction, our analysis contrasts with Board (2011) and Andrews and Barron (2016), and task assignment here functions differently from other canonical incentive instruments examined in prior work.

Lemma 1 also implies that sustaining a first-best equilibrium requires at least two workers, which we assume hereafter. If there were only one worker but a first-best equilibrium existed, then along the path of this equilibrium the worker must be assigned in every period and would always receive zero payoff, violating the incentive constraint.

4 Uniformly first-best assignment

In this section, we report our main result. Definition 2 is essential.

Definition 2 (First-order rotation). *An assignment rule is a first-order rotation rule if it divides the horizon into two phases that operate as follows:*

1. *In the first phase, starting from an initial assignee at time zero, the current assignee is retained after a bad output and relieved after a good output. When relieved, the next assignee is selected from among workers who have not yet been assigned. This phase ends once every worker has been assigned and has produced one good output.*
2. *In the second phase, in each period workers are distinctly labeled $\pi_1, \dots, \pi_n$ and the worker labeled π_n is assigned. The initial labeling is determined by the outcome in the first phase: workers assigned earlier in the first phase receive label π_k with higher k . At the end of each period, labels evolve as follows. After a current bad output, all labels remain unchanged. After a current good output, each worker currently labeled π_k is relabeled $\pi_{(k \bmod n)+1}$ for $k = 1, \dots, n$.*

Therefore, under any first-order rotation rule, the manager rotates the task through all workers, although she has the option to permanently bench some worker, and she does so based only on first-order information—namely the incumbent’s identity and his current output—retaining him after a bad output and relieving him after a good output until all other workers have been assigned and delivered a good output. The order through which the workers are assigned in the first phase can be arbitrarily specified, but the order through which they are assigned in the second phase must follow the realized order in first phase. For ease of reference, we call a worker labeled π_k the k th ranked worker, and higher k corresponds to lower rank.

Despite the simplicity of first-order rotation and the manager’s freedom to condition assignment on the entire history of assignments and outputs, our main result shows that first-order rotation rules are the unique class of assignment rules that are uniformly first-best:

Theorem 1 (Characterization). *An assignment rule is uniformly first-best if and only if it is a first-order rotation rule.*

Therefore, uniformly first-best assignment rules exist by construction. While it is intuitive that first-order rotation rules constitute a first-best equilibrium for certain parameter vectors, namely those in which workers’ shirking gain s is sufficiently close to zero, Theorem 1 implies that no assignment rule outperforms first-order rotation

for sustaining the manager's first-best equilibrium, despite the manager's flexibility; if a first-best assignment rule exists, uniformly so or not, then first-order rotation rules are first-best. Indeed, as we explain further below, for certain parameter vectors, any first-best assignment rule must prescribe first-order rotation.

Effort incentives in all first-best equilibria in which the manager plays a first-order rotation rule are identical and that in any such equilibrium, effort incentives are the same across workers and across time. To see this, note that in any such equilibrium, at every on-path history in the second phase, a worker's continuation payoff depends only on his rank π , and we denote this payoff by $U(\pi)$. By (2), each worker's incentive constraint when assigned on path in the second phase is

$$U(\pi_1) - U(\pi_n) \geq \frac{(1 - \delta)s}{\delta(p - q)}, \quad (4)$$

because a good output moves him to the top rank π_1 while a bad output places him at the bottom rank π_n . This inequality is also each worker's incentive constraint when assigned on path in the first phase. This is because in both phases, an assignee faces the same incentive structure: he is retained after a bad output and relieved after a good output until all other workers have been assigned and delivered a good output. Thus, effort incentives when assigned are identical across workers and across time. Finally, first-order rotation rules differ only in the distribution over the order in which workers are assigned in the first phase, during which they face the same said incentive structure. Therefore all first-order rotation rules provide identical incentives. In particular, the continuation payoff function $U(\pi)$ is the same across all first-best equilibria where the manager plays a first-order rotation rule.

The intuition of Theorem 1 is as follows. For each workforce size, maximal moral hazard that permits the existence of a first-best equilibrium, corresponding to parameter vectors on the boundary of the largest set of first-best parameter vectors, requires extreme incentives to be provided to workers when they are assigned. Because perfect public equilibrium has a recursive structure and workers are symmetric in the stage game, these extreme incentives must be identical across time and workers. The harshest feasible punishment is immediate retention, while the strongest feasible reward is relief until every other worker has been assigned and produced a good output; under the rivalry constraint (3), no stronger reward is possible. Therefore, any first-best assignment rule must take the form of first-order rotation. Since these

incentives are effective when moral hazard is maximal while permitting the existence of a first-best equilibrium, they are also effective for interior first-best parameter vectors where moral hazard is less severe.⁵ Hence, first-order rotation rules are first-best for all first-best parameter vectors simultaneously; that is, they are uniformly first-best.

This intuition highlights that any assignment rule that does not prescribe first-order rotation fails to constitute a first-best equilibrium when moral hazard is sufficiently severe, even if it constitutes one for some parameter vectors. In particular, assignment rules featuring probabilistic rewards and punishments that are fine-tuned to model parameters—despite their prominence in sustaining optimal equilibria in the repeated games literature—are not desirable here, not merely because they are typically parameter-sensitive, but because they provide weaker effort incentives. In standard repeated games, probabilistic transitions are often used to limit time spent in surplus-depleting phases while preserving incentives. In contrast, here, the manager attains her first-best in all on-path continuations in any first-best equilibrium. As a result, probabilistic transitions offer no advantage over deterministic ones and only weaken effort incentives.

First-order rotation rules are intuitive and readily implementable in practice. To our knowledge, however, no prior work has shown that first-order rotation implements optimal dynamic incentives. As discussed in the Introduction, they differ from non-performance-contingent rotation, which fails to address moral hazard in our setting. They also depart from assignment rules featuring favoritism and reliance on relative worker comparison (Board, 2011; Andrews and Barron, 2016). Indeed, they subvert the rivalrous nature of task assignment by disregarding relative worker comparison rather than promoting competition based on it.⁶ Intuitively, in our setting, relative performance is pure noise for motivating the assignee’s effort in each period, and so conditioning assignments on this information weakens effort incentives. This echoes the informativeness principle in monetary contracting (Holmström, 1979, 1982), although our setting is substantially different. First-order rotation rules further differ from assignment rules that treat unassigned workers symmetrically, as in De Clippel et al.

⁵For each interior first-best parameter vector, there exists a first-best assignment rule that does not prescribe first-order rotation; see Appendix B.1 for an example.

⁶For comparison, an example of an assignment rule that uses relative performance is as follows. In each period, the manager computes, for each worker, a score equal to the number of good outputs minus the number of bad outputs previously produced, and assigns the worker with the lowest score, breaking ties uniformly at random.

(2021), under which the current assignee is retained after a bad output and relieved after a good output, with the next assignee drawn uniformly from the remaining pool of workers. This provides a weaker reward than under first-order rotation, since an assignee who is relieved after a good output may still be selected again after the next period with positive probability.

A key step in the proof of Theorem 1 involves characterizing the largest set of first-best parameter vectors $\Omega^*(n)$ for each $n \geq 2$. As described in the Introduction, a challenge in this characterization lies in identifying the extreme points of the set of payoff vectors corresponding to first-best equilibria. These extreme points determine the extremal incentives and, in turn, the necessary and sufficient conditions under which a first-best equilibrium exists, yielding $\Omega^*(n)$. The existing approach, which is effective for two workers, begins by noting that the set of workers' payoffs in first-best equilibria has an interval structure and therefore has two extreme points corresponding to the harshest punishment and the maximal reward for the workers; one can then use these extreme points to determine the parameter values under which these extreme incentives are effective so that a first-best equilibrium exists (Athey and Bagwell, 2001; De Clippel et al., 2021). With more workers, however, this approach is not fruitful and such a characterization has not been obtained in the literature, as the geometry of the set of first-best equilibrium payoff vectors is not well understood. Behaviorally, this reflects the difficulty the manager faces when granting respite to a current assignee. Doing so requires choosing a new assignee, and when the workforce size is at least three so that there are at least two other workers to choose from, this decision is nontrivial: it affects all workers' continuation payoffs through the rivalry constraint and, in turn, their incentive constraints in subsequent periods when they are assigned. To circumvent the difficulty, we simultaneously solve for both the boundary of $\Omega^*(n)$ and the extreme points of the set of first-best equilibrium payoff vectors for parameter vectors lying on this boundary. The boundary then pins down the desired set of parameter vectors.

In the remainder of this section, we sketch this characterization. The following notations are essential. For each workforce size $n \geq 2$ and parameter vector $\omega \in \Omega(n)$, let $E \subseteq \mathbf{R}_+^n$ be the set of workers' equilibrium payoff vectors,⁷ with typical element $U = (U^1, \dots, U^n)$ where U^i denotes worker i 's payoff. Let

⁷Here we omit the manager's payoff. As mentioned, in any first-best equilibrium, after any history (on path), the manager's continuation payoff is equal to $p\bar{v} + (1-p)\underline{v}$.

E^* be the set of first-best equilibrium worker payoff vectors. Standard arguments show that E is compact (Abreu, Pearce and Stacchetti, 1990), and so is E^* , since $E^* = \{U \in E : \sum_{i=1}^n U^i = (n-1)r\}$ by the rivalry constraint (3). We construct a first-best equilibrium payoff vector $U^* := (\underline{U}, U^{(1)}, \dots, U^{(n-1)})$ that is an extreme point of E^* . Here, $\underline{U} := \min_{(U^1, \dots, U^n) \in E^*} U^1$ is the lowest payoff a worker can obtain in any first-best equilibrium; $U^{(1)} := \max_{(U, U^2, \dots, U^n) \in E^*} U^2$ is the highest payoff a worker can obtain conditional on some other worker obtaining $\underline{U}$ in any first-best equilibrium; for each $k = 2, \dots, n-1$, $U^{(k)} := \max_{(U, U^{(1)}, \dots, U^{(k-1)}, U^{k+1}, \dots) \in E^*} U^{k+1}$ is the highest payoff a worker can obtain conditional on k other workers obtaining $\underline{U}, U^{(1)}, \dots, U^{(k-1)}$ in any first-best equilibrium. Finally, define $\bar{U} := \max_{(U^1, \dots, U^n) \in E^*} U^1$ as the highest payoff a worker can obtain in any first-best equilibrium. We write E^* as E_ω^* to stress its dependence on ω .

Fix $n \geq 2$. The set $\Omega^*(n)$ is closed and hence contains its boundary that we denote by $\text{bd}(\Omega^*(n))$, because it consists of parameter vectors for which a first-best equilibrium exists where the incentive constraint (2)—a weak inequality—holds for each worker when assigned on path. To characterize $\Omega^*(n)$, we solve simultaneously for the boundary $\text{bd}(\Omega^*(n))$ and the set E_ω^* for each parameter vector $\omega \in \text{bd}(\Omega^*(n))$. For each such parameter vector, moral hazard is maximally tight while permitting the existence of a first-best equilibrium so that the payoff vector U^* we construct above is decomposed, in the sense of Abreu et al. (1990), as follows:

$$\begin{pmatrix} \underline{U} \\ U^{(1)} \\ \vdots \\ U^{(n-1)} \end{pmatrix} = (1-\delta) \begin{pmatrix} 0 \\ r \\ \vdots \\ r \end{pmatrix} + \delta \left[p \begin{pmatrix} U^{(1)} \\ U^{(2)} \\ \vdots \\ \underline{U} \end{pmatrix} + (1-p) \begin{pmatrix} \underline{U} \\ U^{(1)} \\ \vdots \\ U^{(n-1)} \end{pmatrix} \right]. \quad (5)$$

The first row concerns a currently assigned worker 1, stating that his continuation payoff is $\underline{U}$, with a good output changing this payoff to $U^{(1)}$ and a bad output leaving it unchanged. Intuitively, because moral hazard is maximally tight while permitting the existence of a first-best equilibrium, his incentive constraint (2) binds even when his payoff is $\bar{U}$ upon a good output and is $\underline{U}$ upon a bad output, namely

$$\bar{U} - \underline{U} = \frac{(1-\delta)s}{\delta(p-q)}, \quad (6)$$

and this drives worker 1's current continuation payoff to the lower bound $\underline{U}$. By

definition of $U^{(1)}$, it then follows that $\bar{U} = U^{(1)}$ because in each period, by symmetry of the workers, the assigned worker receives $\bar{U}$; this establishes the first row. The second row concerns an unassigned worker 2 whose current continuation payoff is $U^{(1)}$, stating that a good output by worker 1 changes his payoff to $U^{(2)}$ while a bad output leaves it unchanged. This holds because given worker 1's payoff evolution, any alternative worker 2's payoff evolution makes him worse off, contradicting the definition of $U^{(1)}$. The other rows follow analogously. The system (5) has a unique solution $(V^k)_{k=1}^n$ so that $U^* = (V^k)_{k=1}^n$; we provide the closed-form expression of $(V^k)_{k=1}^n$ in the proof. By symmetry of the workers, the boundary $\text{bd}(\Omega^*(n))$ is characterized by all feasible parameter vectors that satisfy the binding incentive constraint (6), with $\bar{U} = V^1$ and $U^{(1)} = V^2$. Writing V^k as $V^k(n, p, r, \delta)$ to emphasize its dependence on the parameters, the desired set of parameter vectors is therefore

$$\Omega^*(n) = \left\{ \omega \in \Omega(n) : V^2(n, p, r, \delta) - V^1(n, p, r, \delta) \geq \frac{(1 - \delta)s}{\delta(p - q)} \right\}. \quad (7)$$

5 Comparative statics

The inequality characterizing $\Omega^*(n)$ in (7) determines the scope for sustaining first-best equilibria. We rewrite the inequality as

$$S(n, p, q, r, s) := V^2(n, p, r, \delta) - V^1(n, p, r, \delta) - \frac{(1 - \delta)s}{\delta(p - q)} \geq 0. \quad (8)$$

In this section, we study comparative statics regarding $S(n, p, q, r, s)$, which we interpret as the scope for sustaining first-best equilibria. This has direct implications for organizational design. For example, organizations can influence workforce size n through staffing decisions. The assignment-unassignment payoff gap r can be shaped by adjusting off-duty benefits or working conditions—for instance, ensuring that time off from assignment comes with flexible scheduling or fewer administrative obligations. The monitoring of effort p can be improved (while keeping q fixed) by granting assignees discretion or autonomy that improve outcomes if used actively and so is valuable only when effort is exerted. The failure of shirking detection q can be reduced (while keeping p fixed) through measures such as activity logs or progress tracking tools that make disengagement more visible without directly affecting the productivity of effort. Finally, workers' shirking gain s can be reduced by ensuring

that shirking provides only limited comfort, for example by requiring the task to be engagement-heavy.

Note that the inequality in (8) is precisely the assignees' incentive constraint (4) on path in the first-best equilibria where the manager plays a first-order rotation rule. This is because in any such equilibrium, the continuation payoffs $(U(\pi_k))_{k=1}^n$ satisfy the system

$$\begin{pmatrix} U(\pi_n) \\ U(\pi_1) \\ \vdots \\ U(\pi_{n-1}) \end{pmatrix} = (1 - \delta) \begin{pmatrix} 0 \\ r \\ \vdots \\ r \end{pmatrix} + \delta \left[p \begin{pmatrix} U(\pi_1) \\ U(\pi_2) \\ \vdots \\ U(\pi_n) \end{pmatrix} + (1 - p) \begin{pmatrix} U(\pi_n) \\ U(\pi_1) \\ \vdots \\ U(\pi_{n-1}) \end{pmatrix} \right],$$

which coincides with (5), implying that $U(\pi_n) = V^1$ and $U(\pi_1) = V^2$. Indeed, Theorem 1 implies that for each workforce size n , the largest set of first-best parameter vectors $\Omega^*(n)$ is equivalent to the largest set of parameter vectors under which some first-order rotation rule constitutes a first-best equilibrium, or equivalently, all first-order rotation rules constitute first-best equilibria because, as explained, all of them provide identical incentives.

Proposition 1 (The scope for sustaining first-best equilibria). *The following hold.*

1. $S(n, p, q, r, s)$ is strictly increasing in n and r , and strictly decreasing in q and s .
2. There exists $p^* \in (q, 1]$ such that $S(n, p, q, r, s)$ is strictly increasing in p on (q, p^*) and strictly decreasing in p on $[p^*, 1]$. There exists $\underline{r} > 0$ such that $p^* \in (0, 1)$ if $r > \underline{r}$ and $p^* = 1$ otherwise.

Part 1 shows that increasing workforce size, widening the assignment-unassignment payoff gap, improving shirking detection, and reducing shirking gain all expand the scope for sustaining first-best equilibria. Intuitively, in any first-best equilibrium in which the manager plays a first-order rotation rule, workers' effort incentives are stronger if the duration of respite following a good output is longer, workers have stronger preference between being assigned and being unassigned, shirking is less likely to yield a good output, and the gain from shirking is lower. This highlights that our model is unlike standard competitive environments where a larger pool of participants undermines effort incentives (Nalebuff and Stiglitz, 1983; Taylor, 1995),

consistent with our earlier observation that first-order rotation subverts rivalry rather than promoting competition.

Part 2 shows that improving effort detection has, in general, ambiguous effects on the scope for sustaining first-best equilibria and may reduce it unless effort detection is initially poor or the assignment-unassignment payoff gap is small. Intuitively, a marginal improvement of effort detection has two opposing effects. On the one hand, it strengthens effort incentives by making effort more likely to generate a good output. On the other hand, because effort becomes more likely to generate a good output for all workers, it accelerates their rotation across ranks and shortens respite, weakening incentives. The positive effect dominates the negative effect if and only if the assignment-unassignment payoff gap is small and effort detection is initially poor so that rotation is slow. Therefore, although both improvements in effort detection and in shirking detection are information instruments, their effects on the scope for sustaining first-best equilibria are fundamentally different, contrary to the conventional wisdom that better monitoring unambiguously improves dynamic incentives in pure moral-hazard settings (Kandori, 1992).

Taken together, Proposition 1 suggests that organizations can improve the scope of effort incentive provision by maintaining a sufficiently large workforce, structuring unpleasant assignments to be engagement- and responsibility-heavy, with sufficient progress-tracking, while ensuring that unassigned periods offer genuine relief. Granting discretion or autonomy that makes effort more productive may however backfire.

6 Effort-equality tension

A remarkable feature of first-order rotation is its discriminatory nature, highlighting a tension between effort incentives and worker equality. In this section, we formalize this observation. For any history h , we define worker i 's output record in this history as the subsequence of h consisting of the output realizations from the periods in which worker i is assigned. We say that a first-best equilibrium (in which workers play identical strategies on path—exerting effort when assigned) is discriminatory if in this equilibrium, there exists a history on path in which two workers with identical output record have different continuation payoffs; it is nondiscriminatory otherwise.⁸

⁸This definition can be extended for equilibria that are not first-best, but it suffices for our purpose.

Proposition 2 (Strict payoff ranking). *In any first-best equilibrium where the manager plays a first-order rotation rule, $U(\pi_k)$ is strictly decreasing in k ; consequently, the equilibrium is discriminatory.*

Proposition 2 shows that in any first-best equilibrium where the manager plays a first-order rotation rule, higher ranked workers have strictly higher continuation payoffs in each period in the second phase. This is intuitive, as higher ranked workers enjoy longer respite. Thus, in this phase, in every continuation on path, workers' workloads, namely their average discounted frequencies of being assigned, are strictly ordered. In particular, there are continuations that arise with positive probability in which two workers who are symmetric in all payoff-relevant respects—ex ante identical, following identical strategies, and having identical output records—are strictly ranked. For example, such ranking occurs at the end of the first phase even if all workers produce a good output immediately when assigned in that phase. Consequently, the equilibrium is discriminatory.

First-order rotation rules need not be discriminatory until the second assignee is relieved in the first phase. This is because the initial assignee can be drawn uniformly at random and when he is relieved, the second assignee can again be drawn uniformly from those who have not yet been assigned. In the first-best equilibrium corresponding to such a first-order rotation rule, workers who are initially symmetric in all respects have the same ex ante payoff; before the second assignee produces an output in the first phase, the only workers who are symmetric in all respects are those who have yet to be assigned, and these workers have the same expected continuation payoff. Once the second assignee is relieved, his average workload is lower than that of the first assignee even if both of them were relieved following the same output record.

First-best equilibria need not be discriminatory; see Appendix B.2 for an example. Proposition 2 implies that in any nondiscriminatory first-best equilibrium, the assignment rule is necessarily different from first-order rotation, and Theorem 1 implies that this rule fails to be first-best for a range of parameters under which moral hazard is sufficiently severe but first-order rotation continues to constitute first-best equilibria. Therefore, Proposition 2 may be interpreted as highlighting that discrimination can be optimal even among symmetric workers and must be so if the moral hazard problem is sufficiently severe.

In practice, discriminatory worker treatment is often considered detrimental in organizations. While such consideration lies outside our model, we next examine how

the various instruments we have examined in Section 5 affect inequality in first-best equilibria where the manager plays a first-order rotation rule. Rather than constructing a summative measure of inequality across all ranks, we examine the payoff inequality for each rank pair in the second phase; we ignore the first phase because workers' initial order can be arbitrarily specified. Specifically, we define the inequality between any two workers ranked k th and k' th as their (absolute) payoff difference in the first-best equilibrium where the manager plays a first-order rotation rule, namely⁹

$$I(k, k'; n, p, r) := |U(\pi_k; n, p, r) - U(\pi_{k'}; n, p, r)|,$$

where workers' continuation payoff function $U(\cdot)$ is written as $U(\cdot; n, p, r)$ to emphasize its dependence on the parameters; it is independent of q and s because workers do not shirk on path. Note that $I(k, k'; n, p, r)$ is symmetric in k and k' .

Proposition 3 (Comparative statics concerning inequality). *Fix distinct ranks $k \neq k'$. Then:*

1. $I(k, k'; n, p, r)$ is independent of q and s .
2. $I(k, k'; n, p, r)$ is strictly decreasing in n but $I(1, n; n, p, r) = I(n, 1; n, p, r)$ is strictly increasing in n .
3. $I(k, k'; n, p, r)$ is strictly increasing in r .
4. The following hold.
 - (a) If one of k, k' equals n , then $I(k, k'; n, p, r)$ is strictly decreasing in p .
 - (b) If neither k nor k' equals n , there exists $p^\dagger \in (0, 1)$ such that $I(k, k'; n, p, r)$ is strictly increasing in p on $(0, p^\dagger)$ and strictly decreasing in p on $[p^\dagger, 1)$.

Part 1 shows that both shirking detection and shirking gain do not affect worker inequality, because in first-best equilibria workers exert effort when assigned on path. Part 2 shows that increasing workforce size reduces inequality for any fixed rank pair but increases inequality between the top and bottom ranks. This is because a larger workforce lengthens respite after success, thereby increasing the spread between extreme ranks while reducing the marginal gains from moving up the ranking. Part 3 shows that expanding the assignment-unassignment payoff gap increases inequality

⁹Adopting any other measure of inequality that is a positive transformation of this payoff difference does not affect our results.

for any fixed rank pair, as it scales the value of respite across all ranks. Finally, part 4(a) shows that a marginal improvement in effort detection unambiguously reduces inequality for any rank pair involving the bottom rank, whereas part 4(b) shows that for other rank pairs it reduces inequality if and only if effort detection is already sufficiently precise. This is because better effort detection accelerates rank transitions, as discussed, thereby reducing inequality by shortening each worker’s time spent at any given rank. At the same time, it raises workers’ value of being assigned at a later time in the future, which increases inequality. The inequality-reducing effect applies across all ranks, whereas the inequality-increasing effect arises only among non-bottom-ranked workers who are currently unassigned. This establishes part 4(a). For other rank pairs, the inequality-reducing effect dominates if and only if current effort detection is sufficiently precise so that rotation is fast, establishing part 4(b). Figure 1 illustrates.

Taken together, Proposition 1 and Proposition 3 suggest that reducing shirking gain or improving shirking detection is especially attractive for organizations seeking to motivate effort without exacerbating inequality (irrespective of the cost of adopting the various instruments that is not captured in our model). Unlike increasing the workforce size or the assignment-unassignment payoff gap, they strengthen effort incentives without affecting inequality. Their comparison with improving effort detection is more nuanced. For some parameters, improving effort detection can strengthen effort incentives while simultaneously reducing inequality; see Figure 2.¹⁰ Nonetheless, shirking detection has a robustness advantage: improving effort detection may either undermine effort incentives or increase inequality, depending on the parameters, as depicted in Figure 1.

7 Desirable task

Our baseline model assumes that the task is undesirable for workers. In this section, we instead consider a desirable task, so that workers prefer being assigned to being unassigned. We show that all our results extend to this setting in a natural way, with an exception on the comparative statics with respect to the precision p of effort monitoring. Specifically, suppose that in each period, the unassigned worker gets

¹⁰Appendix B.3 shows that this occurs when workers are sufficiently patient and the current quality of effort detection is sufficiently poor.

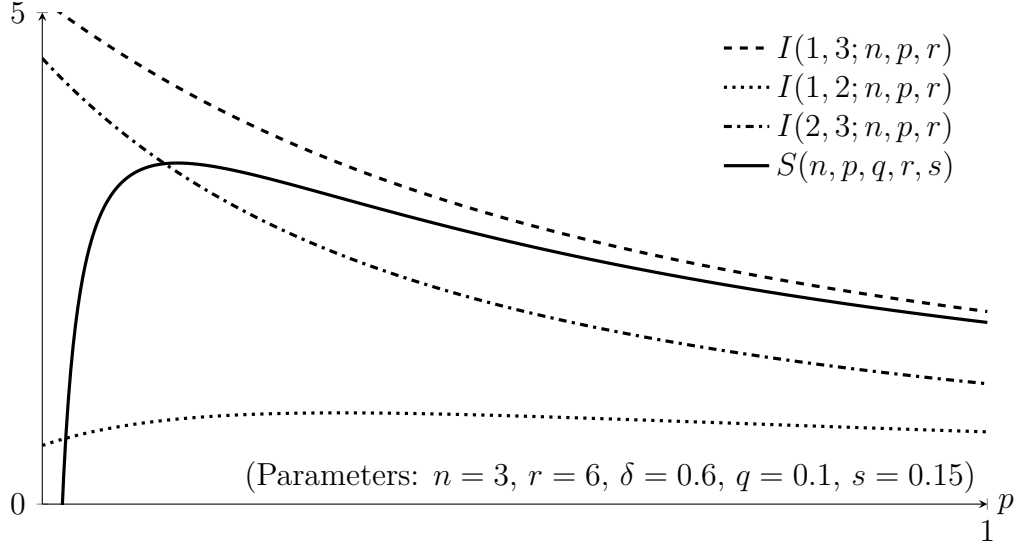


Figure 1: Scope for sustaining first-best equilibria and inequalities

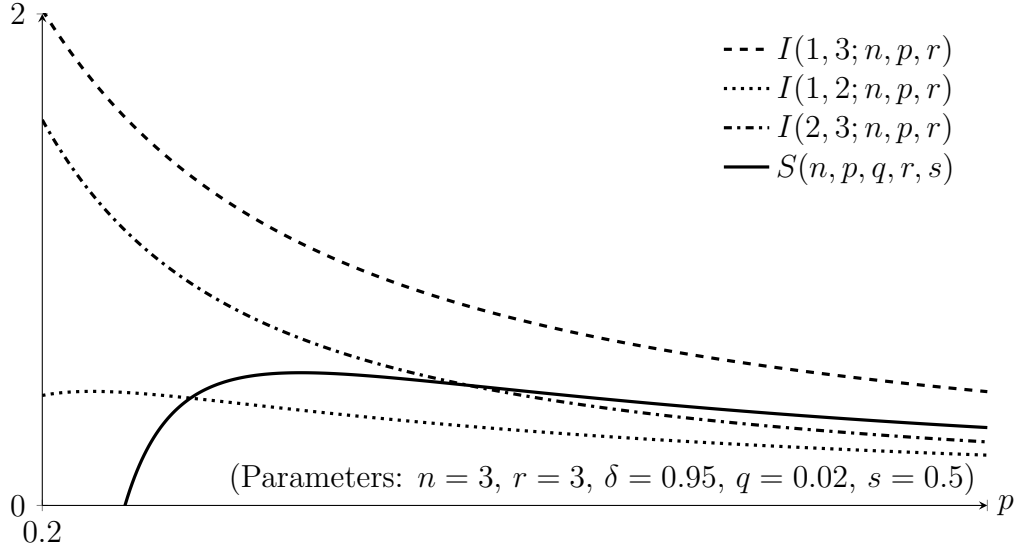


Figure 2: Scope for sustaining first-best equilibria and inequalities

payoff $-r < 0$ instead of r . The model is otherwise unchanged. In particular, upon being assigned, each worker continues to prefer to shirk than to exert effort, and higher r continues to represent stronger worker preference between being assigned and being unassigned.

In this setting, the statements of our results extend, except part 2 of Proposition 1 and part 4(a) of Proposition 3 which concern the comparative statics with respect to effort detection p . The definition of first-order rotation must however be modified so that each worker when assigned is retained upon a good output but relieved upon a bad output until all others have been assigned and produced a bad output. This is because here assignment serves as a reward rather than a punishment to motivate effort, and so the logic of rotation must be flipped.

Definition 3 (First-order rotation with desirable task). *In this desirable-task setting, an assignment rule is a first-order rotation rule if it divides the horizon into two phases that operate as follows:*

1. *In the first phase, starting from an initial assignee at time zero, the current assignee is retained after a good output and relieved after a bad output. When relieved, the next assignee is selected from among workers who have not yet been assigned. This phase ends once every worker has been assigned and has produced one bad output.*
2. *In the second phase, in each period workers are distinctly labeled $\pi_1, \dots, \pi_n$ and the worker labeled π_n is assigned. The initial labeling is determined by the outcome in the first phase: workers assigned earlier in the first phase receive label π_k with higher k . At the end of each period, labels evolve as follows. After a current good output, all labels remain unchanged. After a current bad output, each worker currently labeled π_k is relabeled $\pi_{(k \bmod n)+1}$ for $k = 1, \dots, n$.*

Proposition 4 below states the counterpart of part 2 of Proposition 1 and part 4(a) of Proposition 3 in this setting:

Proposition 4. *In this desirable-task setting:*

1. *$S(n, p, q, r, s)$ is strictly increasing in p .*
2. *For any distinct ranks $k \neq k'$, with one of them equal to n , $I(k, k'; n, p, r)$ is strictly increasing in p .*

Therefore, unlike part 2 of Proposition 1, here improving effort detection unambiguously expands the scope for sustaining first-best equilibria. Intuitively, this improvement has two effects. First, it raises the likelihood that effort produces a good output. Second, it slows down rotation of the task across workers, thereby making the punishment each assignee faces after a bad output more severe. Both effects strengthen effort incentives. On the other hand, unlike part 4(a) of Proposition 3, here a marginal improvement in effort detection unambiguously increases, rather than decreases, inequality for any rank pair involving the bottom rank π_n at which a worker is assigned the task, because this current assignee becomes more likely to stay on the task after exerting effort.

Practically, Proposition 4 highlights that for assignment of a desirable task, unlike that of an undesirable task, improving effort detection would not backfire for expanding the scope for sustaining first-best equilibria. In addition, irrespective of the initial parameter values, such improvement is unambiguously bad for organizations seeking to strengthen effort incentives without exacerbating inequality; it is strictly dominated by instruments that improve shirking detection or those that reduce shirking gain.

8 Extensions

In this section, we explore several variants of our baseline model.

Time-varying preferences. Consider a version of our baseline model in which the workers face private idiosyncratic shocks to their opportunity cost to exert effort. Specifically, suppose that in each period, each worker i has a private shirking gain s^i that is independently drawn across time and workers from some (possibly worker-specific) distribution on a nondegenerate compact interval $[0, s]$. The statements of our results continue to hold. This is because if a first-best equilibrium exists, then at every history, the assignee must have a best reply to exert effort irrespective of his realized shirking gain, including the highest realization s , which is the case in the baseline model.

Multiple tasks. While we have interpreted our model as the manager having a single task and repeatedly assigning it to workers, it can equivalently be viewed as the manager assigning a new but homogeneous task in each period. Under this latter

interpretation, our results extend to a setting where in each period the new task to be assigned may be either desirable or undesirable for the workers. Appendix B.4 shows that our results continue to hold, combining the logic of our undesirable-task baseline model and the desirable-task variant above.

An implicit assumption in this extension is that a new task is assigned only after the previous one is completed. In some applications, however, tasks overlap. This can be captured by allowing the manager to assign an undesirable task and a desirable task concurrently in each period. In this setting, the uniformity criterion is an unrealistic desideratum: no uniformly first-best assignment rule exists. The reason is that in each period, assignment dynamics must now respond jointly to the outputs of both assignees and the monitoring structures of both tasks are in general different. When moral hazard is maximally tight while still permitting the existence of first-best equilibria, incentive provision requires rewards and punishments that make both assignees' incentive constraints bind as in our baseline model, which in turn requires these incentives to be tailored to the monitoring structures, violating the uniformity criterion. Nonetheless, the techniques developed for our baseline model remain applicable. They allow us to characterize the set of first-best parameter vectors and to construct, for each such vector, a first-best assignment rule. Appendix B.5 illustrates this.

Limited worker information. In some applications, such as gig-economy platforms that dispatch tasks, workers may not observe who is assigned a task or the resulting output, except when they themselves are assigned. Consider a variant of our baseline model in which, in each period, unassigned workers observe neither the assignee's identity nor his output; the model is otherwise unchanged. In this setting, it is natural to allow the manager and the workers to employ private strategies, unlike in the baseline model. We accordingly adopt belief-free equilibrium (Ely, Hörner and Olszewski, 2005) as the solution concept, requiring that players' strategies remain their best replies irrespective of their private histories. This solution concept is consistent with the uniformity criterion, as it precludes assignment rules that rely on the manager's precise knowledge of workers' beliefs or on the manager relentlessly tracking workers' belief dynamics. Belief-free equilibrium has a recursive structure, and our results extend following analogous arguments in the baseline model.

Workers may also be uncertain about the true workforce size n . Accordingly,

suppose further that the manager has private information about n , while workers hold a common prior belief on n . This setting features incomplete information and we extend the notion of belief-free equilibrium to Yamamoto’s (2014) ex post belief-free equilibrium, requiring that players’ strategies remain their best replies irrespective of their private histories and the true state n . This solution concept likewise is consistent with the uniformity criterion and preserves a recursive structure, and our results continue to hold. Strictly speaking, both Ely et al. (2005) and Yamamoto (2014) define belief-free equilibria for two-player finite games only, but their definitions can be adopted here in the natural way.

In settings captured by the preceding paragraphs, workers are uncertain about their relative ranks under first-order rotation. This highlights that the effort-equality tension in our model is robust to rank uncertainty, as mentioned earlier, unlike in monetary contracting among complementary workers (Halac et al., 2021).

9 Concluding comments

In this paper, we have examined task assignment among workers as an incentive instrument under dynamic moral hazard and also elucidated novel features of the resulting optimal incentives. While a pure moral hazard setting is a natural point of departure, in some applications workers may have private information about their ability that plays a nonnegligible role in managerial assignments. Consider again our baseline undesirable-task model. If each worker might be a type that is inherently incapable of producing a good output, then a manager who is sufficiently pessimistic about this worker’s ability might choose not to assign him even after he produced a bad output; this undermines the scope of punishment. A systematic analysis of how the workers’ private information shapes assignment dynamics requires distinct technical tools. This is because the recursive equilibrium structure that we have exploited in this paper does not apply and it is not satisfactory to adopt a belief-free solution concept that restores the recursive structure at the expense of muting the manager’s reliance on her belief about the workers in determining assignments. We leave this to future research.

A Proofs

A.1 Proof of Lemma 1

Fix a first-best equilibrium and history h_t on path where worker i is assigned with positive probability. Worker i 's continuation payoff by exerting effort is given by

$$U^i(h_t i) = (1 - \delta) \times 0 + (pU^i(h_t i \bar{y}) + (1 - p)U^i(h_t i \underline{y})).$$

By a one-shot deviation to shirk, this worker's continuation payoff is

$$\hat{U}^i(h_t i) = (1 - \delta) \times s + (qU^i(h_t i \bar{y}) + (1 - q)U^i(h_t i \underline{y})).$$

The one-shot deviation principle requires that $U^i(h_t i) - \hat{U}^i(h_t i) \geq 0$. Rearranging yields (2).

Next, in each period on path in the continuation following history h_t , one worker is assigned and exerts effort while the other $n - 1$ workers are unassigned. Because in each such period the worker who is assigned and exerts effort gets payoff 0 while each unassigned worker gets payoff r , (3) follows.

A.2 Proof of Theorem 1

Fix $n \geq 2$. Recall the definitions of $\underline{U}$, $\bar{U}$, and $(U^{(k)})_{k=1}^n$ in Section 4. We proceed in three steps.

Step 1: Characterizing $\Omega^*(n)$

We show that the set $\Omega^*(n)$ satisfies (7). Define

$$\xi := \frac{1 - \delta(1 - p)}{\delta p}, \tag{9}$$

and

$$V^1(n, p, r, \delta) := \frac{\xi^{n-1} - 1}{\xi^n - 1} r, \quad (10)$$

$$V^k(n, p, r, \delta) := \left(1 - \frac{(\xi - 1)\xi^{k-2}}{\xi^n - 1}\right) r, \quad \text{for } k = 2, \dots, n. \quad (11)$$

We first characterize $\text{bd}(\Omega^*(n))$ and the extreme points of E_ω^* for each $\omega \in \text{bd}(\Omega^*(n))$. To do so, it is without loss of generality to, and we will, assume that the manager plays a pure strategy, because all continuation payoff vectors that take place on path of any first-best equilibrium whose associated payoff vector is an extreme point of $\omega \in \text{bd}(\Omega^*(n))$ must correspond to some equilibrium where the manager plays a pure strategy. Consequently, in any first-best equilibrium, at any history h_t following which worker i is assigned, $U^i(h_t) = U^i(h_{ti})$, where h_{ti} denotes the concatenation of history h_t followed by an assignment of worker i .

We proceed via a series of lemmas. Recall that $\Omega^*(n)$ is closed and so contains its boundary.

Lemma 2. *Let $\omega \in \text{bd}(\Omega^*(n))$. In any first-best equilibrium, at any history h_t following which worker i is assigned,*

$$U^i(h_t) \geq \frac{ps}{p - q}. \quad (12)$$

Proof of Lemma 2. Fix $\omega \in \text{bd}(\Omega^*(n))$ and a first-best equilibrium. By Lemma 1, at any history h_t , the assigned worker i 's incentive constraint for effort holds:

$$U^i(h_{ti}\bar{y}) - U^i(h_{ti}\underline{y}) \geq \frac{(1 - \delta)s}{\delta(p - q)}. \quad (13)$$

Therefore, by definition of $\underline{U}$,

$$\begin{aligned} U^i(h_t) &= U^i(h_{ti}) \\ &= \delta(pU^i(h_{ti}\bar{y}) + (1 - p)U^i(h_{ti}\underline{y})) \geq \delta \left(U^i(h_{ti}\underline{y}) + \frac{(1 - \delta)ps}{\delta(p - q)} \right) \geq \delta \underline{U} + \frac{(1 - \delta)ps}{(p - q)}, \end{aligned}$$

where the first equality follows because the manager plays a pure strategy. Because the first-best equilibrium, the history h_t , and the assigned worker i are arbitrarily picked, the above inequality holds when they are picked such that $U^i(h_t)$ attains $\underline{U}$,

giving

$$\underline{U} \geq \delta \underline{U} + \frac{(1-\delta)ps}{(p-q)} \implies \underline{U} \geq \frac{ps}{(p-q)}. \quad (14)$$

Because $U^i(h_t) \geq \underline{U}$, (14) implies (12), as was to be shown. $\blacksquare$

Lemma 3. *Let $\omega \in \text{bd}(\Omega^*(n))$. It holds that*

$$\bar{U} - \underline{U} = \frac{(1-\delta)s}{\delta(p-q)}. \quad (15)$$

In any first-best equilibrium, for each $i = 1, \dots, n$, at each history where worker i is assigned, upon producing a good output, his continuation payoff is equal to $\bar{U}$; upon producing a bad output, his continuation payoff is $\underline{U}$.

Proof of Lemma 3. Fix $\omega \in \text{bd}(\Omega^*(n))$ and a first-best equilibrium. By Lemma 1, at history h , upon worker i being assigned, $U^i(hiy) \geq \underline{U}$. Next, at any history h' , because there must be one worker who is assigned and then exerts effort, $U^i(h') \leq \bar{U}$. Consequently, $U^i(hi\bar{y}) \leq \bar{U}$. Therefore (2) implies

$$\bar{U} - \underline{U} \geq \frac{(1-\delta)s}{\delta(p-q)}. \quad (16)$$

Note that $\bar{U}$ and $\underline{U}$ depend on ω . To complete the proof, it suffices to show that this inequality cannot be strict. If it were strict, then by continuity in ω on both sides of (16), there exists an open ball B_ω containing ω such that (16) holds and a first-best assignment rule exists given each $\omega' \in B_\omega$, contradicting that $\omega \in \text{bd}(\Omega^*(n))$. $\blacksquare$

Lemma 4. *Let $\omega \in \text{bd}(\Omega^*(n))$. In any first-best equilibrium, for each $i = 1, \dots, n$, at each history h where worker i is assigned, his continuation payoff $U^i(hi)$ is equal to $\underline{U}$.*

Proof of Lemma 4. Fix $\omega \in \text{bd}(\Omega^*(n))$ and a first-best equilibrium, history h , and worker i assigned at this history. At this history, worker i 's continuation payoff is

$$(1-\delta) \times 0 + \delta(p\bar{U} + (1-p)\underline{U}). \quad (17)$$

This is because, by Lemma 3, $U^i(hi\bar{y}) = \bar{U}$ and $U^i(hi\underline{y}) = \underline{U}$. Therefore, (15) and

(17) together imply that

$$U^i(h) = (1 - \delta) \frac{ps}{p - q} + \delta \underline{U}. \quad (18)$$

Because $U^i(h) \geq \underline{U}$ by definition of $\underline{U}$, (18) implies

$$(1 - \delta) \frac{ps}{p - q} + \delta \underline{U} \geq \underline{U} \implies \underline{U} \leq \frac{ps}{p - q}. \quad (19)$$

This and (14) together imply that $\underline{U} = ps/(p - q)$. Consequently, by (18),

$$U^i(h) = (1 - \delta) \frac{ps}{p - q} + \delta \frac{ps}{p - q} = \frac{ps}{p - q} = \underline{U},$$

as desired. ■

Lemma 5. *Let $\omega \in \text{bd}(\Omega^*(n))$. It holds that $\bar{U} = U^{(1)}$.*

Proof. Fix $\omega \in \text{bd}(\Omega^*(n))$. In any first-best equilibrium, in each period, the assigned worker must have continuation payoff $\underline{U}$ by Lemma 4. Therefore, for any $U \in E_\omega^*$, at least one entry in U is equal to $\underline{U}$. Therefore

$$\bar{U} = \max_{(U^1, \dots, U^n) \in E_\omega^*} U^1 = \max_{(U^1, \dots, \underline{U}) \in E_\omega^*} U^1 = U^{(1)},$$

as desired. ■

Lemma 6. *Let $\omega \in \text{bd}(\Omega^*(n))$. Then (5) holds.*

Proof of Lemma 6. Let $\omega \in \text{bd}(\Omega^*(n))$ and $U^* = (\underline{U}, U^{(1)}, \dots, U^{(n-1)}) \in E_\omega^*$. Define $U' := (U^{(1)}, \dots, U^{(n-1)}, \underline{U})$. Because $U^* \in E_\omega^*$, it holds that $U' \in E_\omega^*$ by symmetry. Define also

$$\lambda(y) := \begin{cases} U', & \text{if } y = \bar{y}, \\ U^*, & \text{if } y = \underline{y}. \end{cases}$$

To prove the lemma, we must show that

$$U = (1 - \delta)(0, r, \dots, r) + \delta[p\lambda(\bar{y}) + (1 - p)\lambda(\underline{y})]. \quad (20)$$

Because $U^*, U' \in E_\omega^*$, and because $U^{(1)} - \underline{U} = (1 - \delta)s/[\delta(p - q)]$ by Lemma 3 and Lemma 5, there exists $\hat{U} := (\hat{U}_1, \dots, \hat{U}_n) \in E_\omega^*$ that can be decomposed as follows:

$$\hat{U} = (1 - \delta)(0, r, \dots, r) + \delta[p\lambda(\bar{y}) + (1 - p)\lambda(\underline{y})]. \quad (21)$$

To complete the proof, it suffices to show that $\hat{U} = U^*$. By (21),

$$\hat{U}_1 = \delta(pU^{(1)} + (1 - p)\underline{U}) = \underline{U}, \quad (22)$$

where the second equality uses Lemma 3 and Lemma 4. For any output y , let $\lambda_i(y)$ denote the i th entry of $\lambda(y)$. Because $\lambda_1(\bar{y}) = U^{(1)}$ and $\lambda_1(\underline{y}) = \underline{U}$, it holds that $\lambda_2(\bar{y}) \leq U^{(2)}$ and $\lambda_2(\underline{y}) \leq U^{(1)}$. Because each worker's per-period payoff in any first-best equilibrium is at most r , and because $U^* \in E_\omega^*$,

$$U^{(1)} \leq (1 - \delta)r + \delta(pU^{(2)} + (1 - p)U^{(1)}) = \hat{U}_2 \leq U^{(1)},$$

where the first equality uses (21) and the last inequality uses the definition of $U^{(1)}$. Therefore $\hat{U}_2 = U^{(1)}$. Then, by induction, it holds that for each $k = 2, \dots, n - 2$, given $\hat{U}_1 = \underline{U}$, $\hat{U}_2 = U^{(1)}$, $\dots$, $\hat{U}_k = U^{(k-1)}$,

$$U^{(k)} = (1 - \delta)r + \delta(pU^{(k+1)} + (1 - p)U^{(k)}) = \hat{U}_{k+1} \leq U^{(k)},$$

and therefore $U^{(k)} = \hat{U}_{k+1}$. Finally, because $\lambda(\bar{y}), \lambda(\underline{y}) \in E_\omega^*$, it follows that

$$U^{(n-1)} = (1 - \delta)r + \delta(p\underline{U} + (1 - p)U^{(n-1)}),$$

and therefore $U^{(n-1)} = \hat{U}_n$ by (21). Thus, $\hat{U} = U^*$, as desired. ■

Finally, it is straightforward to verify that for each $\omega \in \text{bd}(\Omega^*(n))$, $U^* = (V^1, \dots, V^n)$, given in (10) and (11), is the unique solution to (5). Therefore $\underline{U} = V^1$ and $U^{(1)} = V^2$, and so

$$\text{bd}(\Omega^*(n)) = \left\{ \omega \in \Omega(n) : V^2(n, p, r, \delta) - V^1(n, p, r, \delta) = \frac{(1 - \delta)s}{\delta(p - q)} \right\}, \quad (23)$$

which implies (7), as desired.

Step 2: First-order rotation rules are uniformly first-best

We next prove that all first-order rotation rules are uniformly first-best. Fix $\omega \in \Omega^*(n)$ and a strategy profile in which each worker exerts effort when assigned and the manager plays some first-order rotation rule. This strategy profile is stationary and symmetric across workers. Under this strategy profile, in each period on path, the continuation payoff of a worker with rank π is denoted by $U(\pi)$. The vector $(U(\pi_n), U(\pi_1), \dots, U(\pi_{n-1}))$ is therefore a unique solution to the system

$$\begin{pmatrix} U(\pi_n) \\ U(\pi_1) \\ \vdots \\ U(\pi_{n-1}) \end{pmatrix} = (1 - \delta) \begin{pmatrix} 0 \\ r \\ \vdots \\ r \end{pmatrix} + \delta \left[p \begin{pmatrix} U(\pi_1) \\ U(\pi_2) \\ \vdots \\ U(\pi_n) \end{pmatrix} + (1 - p) \begin{pmatrix} U(\pi_n) \\ U(\pi_1) \\ \vdots \\ U(\pi_{n-1}) \end{pmatrix} \right]. \quad (24)$$

Because $(V^1, \dots, V^n)$, given in (10) and (11), is the unique solution to (5), and because (5) coincides with (24), it follows that

$$(U(\pi_n), U(\pi_1), \dots, U(\pi_{n-1})) = (V^1, V^2, \dots, V^{n-1}, V^n).$$

Therefore,

$$U(\pi_1) - U(\pi_n) = V^2 - V^1 \geq \frac{(1 - \delta)s}{\delta(p - q)},$$

where the inequality uses the characterization of $\Omega^*(n)$ in Step 1. Consequently, the strategy profile constitutes an equilibrium. This equilibrium is first-best because workers exert effort when assigned on path. Because the first-order rotation rule is arbitrarily chosen, first-order rotation rules are first-best. Because first-order rotation rules do not depend on ω , the above arguments hold simultaneously for all $\omega \in \Omega^*(n)$. Therefore first-order rotation rules are uniformly first-best.

Step 3: Uniform first-best implies first-order rotation

Finally, we prove that all uniformly first-best assignment rules must prescribe first-order rotation. It suffices to show that for any $\omega \in \text{bd}(\Omega^*(n))$, all first-best assignment rules must prescribe first-order rotation. Fix any $\omega \in \text{bd}(\Omega^*(n))$ and any first-best equilibrium. The arguments in the first step imply that in this equilibrium, in each

period on path, the assigned worker's, say worker 1's, continuation payoff is $\underline{U}$; a good output raises this payoff to $U^{(1)}$, whereas a bad output leaves it unchanged. Once worker 1's continuation payoff reaches $U^{(1)}$, he is unassigned and some other worker, say worker 2, is assigned with continuation payoff $\underline{U}$. If there are only two workers, then a good output by worker 2 transitions their payoffs from $(U^{(1)}, \underline{U})$ to $(\underline{U}, U^{(1)})$, while a bad output leaves them unchanged. The transition for worker 2's payoff follows the logic described above, whereas the transition for worker 1's payoff follows because, given that worker 2 is granted respite, worker 1 must be assigned. If there are more than two workers, then a good output by worker 2 transitions their payoffs from $(U^{(1)}, \underline{U})$ to $(U^{(2)}, U^{(1)})$, while a bad output leaves them unchanged. The transition for worker 2's payoff follows the logic described above, whereas the transition for worker 1's payoff holds because, given worker 2's payoff transition, any alternative payoff transition for worker 1 would make him worse off, contradicting that $U^{(1)}$ is the highest continuation payoff a worker can obtain in any first-best equilibrium. When these workers hold continuation payoffs $(U^{(2)}, U^{(1)})$, they are unassigned and a third worker must be assigned with continuation payoff $\underline{U}$. Iterating this argument shows that for each worker, conditional on currently being assigned, the continuation payoff starts at $\underline{U}$, moves cyclically through $U^{(1)}, \dots, U^{(n-1)}$ before returning to $\underline{U}$, changing value after a good output but staying put after a bad output. Consequently, in any first-best equilibrium, a worker who is currently assigned and then produces a good output rests until each other worker has been assigned and produced a good output. The manager's strategy must therefore prescribe first-order rotation.

The three steps together imply Theorem 1, completing the proof.

A.3 Proof of Proposition 1

From the proof of Theorem 1, $U(\pi_k) = V^{k+1}$ and $U(\pi_n) = V^1$. In this proof, we sometimes write ξ as $\xi(p)$ to emphasize its dependence on p . By (10) and (11),

$$S(n, p, q, r, s) = r \frac{(\xi(p) - 1)(\xi(p)^{n-1} - 1)}{\xi(p)^n - 1} - \frac{(1 - \delta)s}{\delta(p - q)}. \quad (25)$$

Part 1 is immediate from (25).

Here we prove part 2. In this proof, we write $S(n, p, q, r, s)$, given in (25), as $S(p)$ to stress its dependence on p . We show that S admits a unique maximizer on $(q, 1]$.

Let $M(\xi) := \frac{(\xi-1)(\xi^{n-1}-1)}{\xi^n-1}$ and $C := s(1-\delta)/\delta$. We can then rewrite (25) as

$$S(p) = rM(\xi(p)) - \frac{C}{p-q}.$$

Differentiating,

$$S'(p) = rM'(\xi)\xi'(p) + \frac{C}{(p-q)^2}.$$

Using $\xi'(p) = -\delta(\xi-1)^2/(1-\delta)$ and the definition of ξ , the equation $S'(p) = 0$ is equivalent to

$$rM'(\xi) = \frac{s}{(1-K\kappa)^2},$$

where $K := \delta q/(1-\delta) > 0$ and $\kappa := \xi - 1 > 0$. The above equation can then be written as

$$K\kappa + \sqrt{\frac{s}{r}} \frac{1}{\sqrt{M'(\kappa+1)}} = 1.$$

Define the left side of this equation as $G(\kappa)$. The number of solutions to $S'(p) = 0$ is exactly the number of solutions to $G(\kappa) = 1$. Note that the function M is strictly concave for all $\xi > 1$ and $n \geq 2$, and so M' is strictly decreasing. Therefore G is strictly increasing and can cross the value of 1 at most once. This proves that S' has at most one root on $(q, 1)$. Furthermore, as $p \searrow q$, $S'(p) \rightarrow \infty$. Therefore, either there exists $p^* \in (q, 1)$ such that S strictly increases to this interior peak p^* and then strictly decreases on $(p^*, 1)$, or S is strictly increasing in $p \in (q, 1]$ and so S is maximized at $p^* = 1$. Note that

$$S'(1) = -r \left(\frac{\delta(1-\delta)(1-\delta^{n-1})^2 + (n-1)(1-\delta)^3\delta^{n-1}}{(1-\delta^n)^2} \right) + \frac{(1-\delta)s}{\delta(1-q)^2}.$$

Define $\underline{r}$ by

$$\underline{r} := \frac{s}{(1-q)^2} \frac{(1-\delta^n)^2}{\delta^2(1-\delta^{n-1})^2 + (n-1)(1-\delta) + \delta^n}.$$

Then, if $r \geq \underline{r}$, $S'(1) < 0$, implying that S admits a unique interior peak $p^* \in (q, 1)$.

Otherwise, S is strictly increasing on $(q, 1]$ and is maximized at $p^* = 1$.

A.4 Proof of Proposition 2

From the proof of Theorem 1, $U(\pi_k) = V^{k+1}$ and $U(\pi_n) = V^1$. By (10) and (11), it follows from direct computation that $V^2 > V^3 > \dots V^n > V^1$, leading to Proposition 2.

A.5 Proof of Proposition 3

In this proof, we sometimes write ξ as $\xi(p)$ to emphasize its dependence on p . By (10) and (11),

$$I(k, k'; n, p, r) = \begin{cases} r \frac{(\xi(p) - 1) (\xi(p)^{n-1} - \xi(p)^{k-2})}{(\xi(p)^n - 1)}, & \text{if } k' = n \text{ and } k < n, \\ r \frac{(\xi(p) - 1) (\xi(p)^{k'-2} - \xi(p)^{k-2})}{(\xi(p)^n - 1)}, & \text{if } n > k' > k, \end{cases} \quad (26)$$

where we have ignored the case $k = n$ and also the case $k > k'$ because $I(k, k'; n, p, r)$ is symmetric in k and k' . Parts 1—3 are immediate. We prove part 4 below. Fix a distinct pair (k, k') . Without loss of generality, let $k' > k$. Write (26) as $I(\xi)$ to emphasize its dependence on ξ . We prove the two parts in order.

- (a). Let $k' = n$. Then, by direct calculations, $I'(\xi) > 0$ if and only if $J(\xi) := \xi^{n-k+2} + (n-k)(\xi-1) - 1 > 0$. This latter inequality holds because $\xi > 1$, $J' > 0$, and $J(1) = 0$. Because $\xi'(p) < 0$ by (9), part (a) follows.
- (b). Let $k' < n$, $k < k'$, and $a := k' - k \in \{1, \dots, n-2\}$. Because $I(\xi) > 0$, the sign of $I'(\xi)$ equals the sign of the log-derivative of I . Let $\phi(\xi) := \log I(\xi)$. By direct calculations,

$$\phi'(\xi) = \frac{1}{\xi-1} + \frac{k-2}{\xi} + \frac{a \xi^{a-1}}{\xi^a-1} - \frac{n \xi^{n-1}}{\xi^n-1}. \quad (27)$$

For $m = 1, 2, \dots$, define

$$g_m(\xi) := \frac{m \xi^{m-1}}{\xi^m-1} = \frac{m}{\xi} \cdot \frac{1}{1-\xi^{-m}} = \frac{m}{\xi} \sum_{\ell=0}^{\infty} \xi^{-m\ell}.$$

Then from (27),

$$\phi'(\xi) = \frac{1}{\xi - 1} + \frac{k - 2}{\xi} + g_a(\xi) - g_n(\xi). \quad (28)$$

Because $\xi'(p) < 0$ by (9), to complete the proof, it suffices to show that $\phi'' < 0$, $\lim_{\xi \searrow 1} \phi'(\xi) = +\infty$, and $\phi'(\xi) \sim \hat{\phi}$ for some $\hat{\phi} < 0$ as $\xi \rightarrow \infty$, where $\sim$ denotes asymptotic equivalence. By (28),

$$\begin{aligned} \phi''(\xi) &= -\frac{1}{(\xi - 1)^2} - \frac{k - 2}{\xi^2} + (g_a - g_n)'(\xi) \\ &< (g_a - g_n)'(\xi) \\ &= \sum_{\ell=0}^{\infty} [-a(a\ell + 1)\xi^{-(a\ell+2)} + n(n\ell + 1)\xi^{-(n\ell+2)}] \\ &< 0. \end{aligned}$$

Next, it is immediate to see from (27) that $\lim_{\xi \searrow 1} \phi'(\xi) = +\infty$. Finally, as $\xi \rightarrow \infty$, $g_m(\xi) \sim m/\xi$ and $1/(\xi - 1) \sim 1/\xi$. Hence, by (27),

$$\phi'(\xi) \sim \frac{1}{\xi} + \frac{k - 2}{\xi} + \frac{a}{\xi} - \frac{n}{\xi} = \frac{k + a - n - 1}{\xi} = \frac{k' - n - 1}{\xi} < 0,$$

as desired. This proves part (b).

A.6 Proof of Proposition 4

We first modify the proof of Theorem 1 to derive S and I . In the first step where we characterize the largest set of first-best parameter vectors $\Omega^*(n)$ given workforce size n , here, unlike in the baseline model, we directly define $U^{(1)}$ as the highest continuation payoff that a worker can obtain in any first-best equilibrium: $U^{(1)} := \max_{(U^1, \dots, U^n) \in E^*} U^1$. Then, iterating for $k = 2, \dots, n$, we define the highest continuation payoff a worker can obtain in any first-best equilibrium conditional on there being k other workers each with continuation payoff $U^{(1)}, \dots, U^{(k-1)}$ by $U^{(k)} := \max_{(U^{(1)}, \dots, U^{(k-1)}, U^k, \dots) \in E^*} U^k$. By construction, both $(U^{(1)}, U^{(2)}, \dots, U^{(n)})$ and $(U^{(n)}, U^{(1)}, \dots, U^{(n-1)})$ are first-best equilibrium payoff vectors. For the same reason as in (5), for each $\omega \in \text{bd}(\Omega^*(n))$,

these vectors satisfy

$$\begin{pmatrix} U^{(1)} \\ U^{(2)} \\ \vdots \\ U^{(n)} \end{pmatrix} = (1 - \delta) \begin{pmatrix} 0 \\ -r \\ \vdots \\ -r \end{pmatrix} + \delta \left[p \begin{pmatrix} U^{(1)} \\ U^{(2)} \\ \vdots \\ U^{(n)} \end{pmatrix} + (1 - p) \begin{pmatrix} U^{(n)} \\ U^{(1)} \\ \vdots \\ U^{(n-1)} \end{pmatrix} \right]. \quad (29)$$

Writing $(\hat{V}^k(n, p, r, \delta))_{k=1}^n$ as the value of $(U^{(k)})_{k=1}^n$ that solves the system (29), the same arguments in the proof of Theorem 1 in the baseline model imply

$$\Omega^*(n) = \left\{ \omega \in \Omega(n) : \hat{V}^1(n, p, r, \delta) - \hat{V}^n(n, p, r, \delta) \geq \frac{(1 - \delta)s}{\delta(p - q)} \right\}.$$

The second and last steps are virtually identical.

Noting that for each $k = 1, \dots, n$,

$$\hat{V}^k(n, p, r, \delta) = -r + \frac{(1 - \delta)r}{1 - \delta p} \cdot \frac{\left(\frac{1 - \delta p}{\delta(1 - p)}\right)^{n-k+1}}{\left(\frac{1 - \delta p}{\delta(1 - p)}\right)^n - 1},$$

it is straightforward to verify that the scope of sustaining first-best equilibria

$$S(n, p, q, r, s) := \hat{V}^1(n, p, r, \delta) - \hat{V}^n(n, p, r, \delta) - \frac{(1 - \delta)s}{\delta(p - q)}$$

is strictly increasing in p , and that, for each $k \neq n$,

$$I(k, n; n, p, r) = I(n, k; n, p, r) = |\hat{V}^1(n, p, r, \delta) - \hat{V}^k(n, p, r, \delta)|$$

is also strictly increasing in p . Consequently, both parts of the proposition follow.

B Omitted details

B.1 Example of probabilistic reward

In this Appendix, we provide an example of an assignment rule that relies on probabilistic transitions and is first-best for a range of parameter vectors but not uniformly first-best. For simplicity, we consider the case of two workers.

Consider an assignment rule where worker 1 is initially assigned. In each period, if the assigned worker delivers a good output, then in the next period the manager assigns the other worker with some probability ζ , chosen below to be such that the currently assigned worker is indifferent between exerting effort and shirking in the corresponding first-best equilibrium, and retains the currently assigned worker otherwise. If the assigned worker delivers a bad output, then he is retained. Fix a strategy profile where the manager plays this assignment rule and the workers exert effort when assigned. By symmetry, a worker's continuation payoff (on path) depends only on whether he is assigned or not, but does not depend on his identity, nor other details of the history of play. Accordingly, write $\hat{U}_S$ as the assigned worker's continuation payoff and $\hat{U}_R$ as the unassigned worker's continuation payoff under this strategy profile. The payoffs $(\hat{U}_S, \hat{U}_R)$ and the transition probability ζ jointly solve the system of promise-keeping constraints

$$\begin{aligned}\zeta(\hat{U}_R - \hat{U}_S) &= \frac{(1 - \delta)s}{\delta(p - q)}, \\ \hat{U}_S &= \delta(p\zeta\hat{U}_R + (1 - p\zeta)\hat{U}_S), \\ \hat{U}_R &= (1 - \delta)r + \delta(p\zeta\hat{U}_S + (1 - p\zeta)\hat{U}_R).\end{aligned}$$

The first equation is the assigned worker's incentive constraint. It is binding because the transition probability upon a good output is chosen to be sufficiently low so that the expected reward from a good output is just enough to make the worker indifferent between exerting effort and shirking whenever assigned. The second and third equations are the workers' promise-keeping constraints. It can be readily verified that when the shirking gain s is sufficiently small, there is a solution to this system, so that the assignment rule constitutes a first-best equilibrium. Nonetheless, in this equilibrium, the transition probability ζ lies in $(0, 1)$ and depends non-trivially on the parameter vector ω for each ω on the interior of $\Omega^*(2)$. Consequently, this assignment rule, while first-best for each such ω , is not uniformly first-best.

B.2 Nondiscriminatory first-best equilibrium

In this section, we present a first-best equilibrium that is nondiscriminatory. For simplicity, we consider the case of two workers. Assume that

$$\frac{r}{2(1 - \delta(1 - p))} \geq \frac{s}{\delta(p - q)}. \quad (30)$$

We construct an equilibrium that depends on a public automaton-state $z = (z_1, z_2) \in \{\emptyset, \bar{y}, \underline{y}\}^2$, where z_i records worker i 's most recent realized output in a period in which i was assigned, with $\emptyset$ meaning “never assigned yet”. As in the main text, we describe on-path behavior only. The initial state is $(\emptyset, \emptyset)$, and after a period in which i is assigned and output $y \in \{\bar{y}, \underline{y}\}$ is realized, the next state replaces z_i by y , leaving z_{-i} unchanged. Given $z = (z_1, z_2)$, the manager's assignment is as follows. If $z_1 = z_2$, then the manager assigns one of the workers with equal probability. If $z \in \{(\emptyset, \bar{y}), (\bar{y}, \emptyset)\}$, then the manager assigns worker i whose $z_i = \emptyset$. Finally, if $z \in \{(\emptyset, \underline{y}), (\underline{y}, \emptyset), (\bar{y}, \underline{y}), (\underline{y}, \bar{y})\}$, the manager assigns worker i whose $z_i = \underline{y}$. Each worker, when assigned, exerts effort.

We verify that this assignment rule is first-best. Given any strategy profile in which the manager plays this rule and workers exert effort when assigned on path, each worker i 's payoff in each continuation on path are functions of the state z ; we abuse notation by writing it as $U^i(z)$ here. Worker 1's continuation payoff at each state z satisfy

$$\begin{aligned} U^1(\underline{y}, \bar{y}) &= U^1(\emptyset, \bar{y}) = \delta(pU^1(\bar{y}, \bar{y}) + (1 - p)U^1(\underline{y}, \bar{y})), \\ U^1(\bar{y}, \underline{y}) &= U^1(\bar{y}, \emptyset) = (1 - \delta)r + \delta(pU^1(\bar{y}, \bar{y}) + (1 - p)U^1(\bar{y}, \underline{y})), \\ U^1(\bar{y}, \bar{y}) &= \frac{1}{2}(\delta(pU^1(\bar{y}, \bar{y}) + (1 - p)U^1(\underline{y}, \bar{y}))) + \frac{1}{2}((1 - \delta)r + \delta(pU^1(\bar{y}, \bar{y}) + (1 - p)U^1(\bar{y}, \underline{y}))), \\ U^1(\underline{y}, \underline{y}) &= \frac{1}{2}(\delta(pU^1(\bar{y}, \underline{y}) + (1 - p)U^1(\underline{y}, \underline{y}))) + \frac{1}{2}((1 - \delta)r + \delta(pU^1(\underline{y}, \bar{y}) + (1 - p)U^1(\underline{y}, \underline{y}))), \\ U^1(\emptyset, \emptyset) &= \frac{1}{2}(\delta(pU^1(\bar{y}, \emptyset) + (1 - p)U^1(\underline{y}, \emptyset))) + \frac{1}{2}((1 - \delta)r + \delta(pU^1(\emptyset, \bar{y}) + (1 - p)U^1(\emptyset, \underline{y}))). \end{aligned}$$

Solving yields

$$\begin{aligned} U^1(\bar{y}, \bar{y}) - U^1(\underline{y}, \bar{y}) &= U^1(\bar{y}, \underline{y}) - U^1(\underline{y}, \underline{y}) = U^1(\bar{y}, \emptyset) - U^1(\underline{y}, \emptyset) \\ &= \frac{(1 - \delta)r}{2(1 - \delta(1 - p))}. \end{aligned}$$

Note then that the left side of worker 1's incentive constraint (2) for effort at any history on path is equal to the above expression. By (30), this incentive constraint holds. By symmetry, worker 2's incentive constraint also holds at any history on path. Thus, the present strategy profile is a first-best equilibrium.

In this equilibrium, it is clear that for every history h , if workers 1 and 2 have identical output record, then their continuation payoffs are identical. This is because they must then have the same output in their most recent assignment, and so both entries in the state associated with that history are identical. Therefore the equilibrium is nondiscriminatory.

B.3 Effort detection on first-best and inequality

In this section, we provide a sufficient condition on the parameters under which a marginal improvement in effort detection improves the scope for sustaining first-best equilibria and simultaneously reduces inequality.

Proposition 5. *For some $\bar{p} > 0$, there exists $\underline{\delta} \in (0, 1)$ such that for each $\delta \geq \underline{\delta}$, $S(n, p, q, r, s)$ is strictly increasing in p on $(q, \bar{p})$ and $I(\cdot; n, p, r)$ is strictly decreasing in p on $(q, \bar{p})$.*

Proof of Proposition 5. As shown in the proof of Proposition 3, as $p \searrow q$,

$$\frac{\partial S(n, p, q, r, s)}{\partial p} \rightarrow \infty.$$

Consequently, for each $\delta \in (0, 1)$, S is strictly increasing in p in some neighborhood of q . Next, by part 4 of Proposition 3, for any distinct k, k' , $I(k, k'; n, p, r)$ is strictly decreasing in p if one of k or k' is equal to n . Finally, for each distinct k, k' with $k' > k$ and neither of them being equal to n , define

$$f_I(\delta) := \lim_{p \searrow q} \frac{\partial I(k, k'; n, p, r)}{\partial p}.$$

By direct calculations,

$$\lim_{\delta \rightarrow 1} f_I(\delta) = 0, \quad \lim_{\delta \rightarrow 1} f'_I(\delta) = \frac{(k' - k)r}{nq^2} > 0.$$

This shows that for each q , when δ is sufficiently close to 1, $f_I(\delta) < 0$. By continuity, there exists $\underline{\delta} \in (0, 1)$ such that $f_I(\delta) < 0$ and so $I(k, k'; n, p, r)$ is strictly decreasing in p in a neighborhood of q . ■

B.4 Multiple tasks

In this section, we provide the omitted details regarding the multiple-task extension in Section 8. Suppose that in each period, unassigned workers receive the same payoff $r_+ > 0$ with probability $\gamma \in (0, 1)$ and $r_- < 0$ with complementary probability. The former event captures that the current task is undesirable whereas the latter captures that it is desirable. In the former event, effort leads to a good output with probability $p_+ \in (0, 1)$ and a bad output with complementary probability, whereas shirking leads to a good output with probability $q_+ \in (0, p_+)$ and a bad output with complementary probability. In the latter, effort leads to a good output with probability $p_- \in (0, 1)$ and a bad output with complementary probability, whereas shirking leads to a good output with probability $q_- \in (0, p_-)$ and a bad output with complementary probability. The assigned worker continues to get payoff 0 if he exerts effort and s if he shirks. The model is otherwise unchanged. The incentive constraint (2) is amended in the obvious way.

As mentioned in the main text, our results extend following a modification of the definition of first-order rotation.

Definition 4 (First-order rotation). *In this setting, an assignment rule is a first-order rotation rule if it divides the horizon into two phases that operate as follows:*

1. *In the first phase, starting from an initial assignee at time zero, the current assignee for an undesirable (resp., desirable) task is retained after a bad (resp., good) output and relieved after a good (resp., bad) output. When relieved, the next assignee is selected from among workers who have not yet been assigned. This phase ends once every worker has been assigned and has produced one good (resp., bad) output.*

2. In the second phase, in each period workers are distinctly labeled $\pi_1, \dots, \pi_n$ and the worker labeled π_n is assigned. The initial labeling is determined by the outcome in the first phase: workers assigned earlier in the first phase receive label π_k with higher k . Thereafter, labels evolve as follows. Given a current undesirable (resp., desirable) task, after a bad (resp., good) output, all labels remain unchanged. After a good (resp., bad) output, each worker currently labeled π_k is relabeled $\pi_{(k \bmod n)+1}$ for $k = 1, \dots, n$.

Extending Theorem 1 requires minimal modifications in the proof. Here we point out the modifications. In the first step where we characterize $\Omega^*(n)$, we define, unlike in the baseline model, $U^{(1)}$ directly as the highest continuation payoff that a worker can obtain in any first-best equilibrium, namely $U^{(1)} := \max_{(U^1, \dots, U^n) \in E^*} U^1$. Then, iterating for $k = 2, \dots, n$, define the highest continuation payoff a worker can obtain in any first-best equilibrium conditional on there being k other workers each with continuation payoff $U^{(1)}, \dots, U^{(k-1)}$ by $U^{(k)} := \max_{(U^{(1)}, \dots, U^{(k-1)}, U^k, \dots) \in E^*} U^k$. By construction, both $(U^{(1)}, U^{(2)}, \dots, U^{(n)})$ and $(U^{(n)}, U^{(1)}, \dots, U^{(n-1)})$ are first-best equilibrium payoff vectors. Further for each $k = 1, \dots, n$, $U^{(k)}$ is an expected continuation payoff, with the expectation taken over the current task desirability, so that it can be written as

$$U^{(k)} = \gamma U_+^{(k)} + (1 - \gamma) U_-^{(k)}, \quad (31)$$

with $U_+^{(k)}$ being the continuation payoff conditional on the current task being undesirable and $U_-^{(k)}$ being the counterpart conditional on the current task being desirable. Then, for each $\omega \in \text{bd}(\Omega^*(n))$, as in (5),

$$\begin{pmatrix} U_+^{(n)} \\ U_+^{(1)} \\ \vdots \\ U_+^{(n-1)} \end{pmatrix} = (1 - \delta) \begin{pmatrix} 0 \\ r_+ \\ \vdots \\ r_+ \end{pmatrix} + \delta \left[p_+ \begin{pmatrix} U^{(1)} \\ U^{(2)} \\ \vdots \\ U^{(n)} \end{pmatrix} + (1 - p_+) \begin{pmatrix} U^{(n)} \\ U^{(1)} \\ \vdots \\ U^{(n-1)} \end{pmatrix} \right], \quad (32)$$

and

$$\begin{pmatrix} U_-^{(1)} \\ U_-^{(2)} \\ \vdots \\ U_-^{(n)} \end{pmatrix} = (1 - \delta) \begin{pmatrix} 0 \\ r_- \\ \vdots \\ r_- \end{pmatrix} + \delta \left[p_- \begin{pmatrix} U^{(1)} \\ U^{(2)} \\ \vdots \\ U^{(n)} \end{pmatrix} + (1 - p_-) \begin{pmatrix} U^{(n)} \\ U^{(1)} \\ \vdots \\ U^{(n-1)} \end{pmatrix} \right]. \quad (33)$$

Writing $(\tilde{V}^k)_{k=1}^n$ as the value of $(U^{(k)})_{k=1}^n$ that solves (31)—(33), the same arguments in the proof of Theorem 1 in the baseline model imply that $\Omega^*(n)$ is the set of $\omega \in \Omega(n)$ satisfying

$$\tilde{V}^1(n, p_+, p_-, r_+, r_-, \delta) - \tilde{V}^n(n, p_+, p_-, r_+, r_-, \delta) \geq \frac{(1 - \delta)s}{\delta \min(p_+ - q_+, p_- - q_-)},$$

where the minimum operator accounts for the fact that the incentive constraint for effort holds irrespective of whether each worker is assigned an undesirable task or a desirable task. The second and last steps are virtually identical. It is then straightforward to verify that the results in the main text carry over.

B.5 Concurrent assignments

In this section, we provide the omitted details regarding the concurrent-assignment extension in Section 8. For simplicity, we consider the case of three workers. Suppose that in each period, the unassigned worker gets payoff $r > 0$, the worker assigned the undesirable task gets payoff 0 if he exerts effort and s if he shirks, and the worker assigned the desirable task gets payoff $x \in (0, r)$ if he exerts effort and $x + s$ if he shirks. When assigned an undesirable task, each worker's effort leads to a good output with probability $p_U \in (0, 1)$ and a bad output with complementary probability, whereas shirking leads to a good output with probability $q_U \in (0, p_U)$ and a bad output with complementary probability. Define (p_D, q_D) as the counterparts for each worker when assigned a desirable task. The model is otherwise identical.

The incentive constraint (2) in Lemma 1 must be modified because two outputs are produced in each period. In any first-best equilibrium, at each history h_t where worker i is assigned the undesirable task and worker j is assigned the desirable task, the incentive constraint for worker i is now

$$\begin{aligned} p_D(U^i(h_t(i, j)(\bar{y}, \bar{y})) - U^i(h_t(i, j)(\underline{y}, \bar{y}))) \\ + (1 - p_D)(U^i(h_t(i, j)(\bar{y}, \underline{y})) - U^i(h_t(i, j)(\underline{y}, \underline{y}))) \geq \frac{(1 - \delta)s}{\delta(p_U - q_U)}, \end{aligned} \quad (34)$$

$$\begin{aligned} p_U(U^j(h_t(i, j)(\bar{y}, \bar{y})) - U^j(h_t(i, j)(\bar{y}, \underline{y}))) \\ + (1 - p_U)(U^j(h_t(i, j)(\underline{y}, \bar{y})) - U^j(h_t(i, j)(\underline{y}, \underline{y}))) \geq \frac{(1 - \delta)s}{\delta(p_D - q_D)}. \end{aligned} \quad (35)$$

where $U^i(h_t(i, j)(y_t^i, y_t^j))$ denotes worker i 's continuation payoff following output y_t^i from worker i and y_t^j from worker j at that history in the equilibrium.

Here, for each workforce size n , a typical parameter vector is $\omega = (p_U, p_D, q_U, q_D, x, r, s, \delta)$. Abusing notation, let $(V^1, V^2, V^3, \bar{\alpha}, \underline{\alpha})$ be the unique solution that solves

$$\begin{aligned} \begin{pmatrix} V^1 \\ V^2 \\ V^3 \end{pmatrix} &= (1 - \delta) \begin{pmatrix} 0 \\ x \\ r \end{pmatrix} + \delta \left[p_D p_U \begin{pmatrix} \bar{\alpha} V^2 + (1 - \bar{\alpha}) V^3 \\ (1 - \bar{\alpha}) V^2 + \bar{\alpha} V^3 \\ V^1 \end{pmatrix} + p_D (1 - p_U) \begin{pmatrix} V^1 \\ V^2 \\ V^3 \end{pmatrix} \right. \\ &\quad \left. + (1 - p_D) p_U \begin{pmatrix} V^2 \\ V^1 \\ V^3 \end{pmatrix} + (1 - p_U)(1 - p_D) \begin{pmatrix} \underline{\alpha} V^3 + (1 - \underline{\alpha}) V^1 \\ (1 - \underline{\alpha}) V^3 + \underline{\alpha} V^1 \\ V^2 \end{pmatrix} \right], \end{aligned} \quad (36)$$

and

$$p_D (\bar{\alpha} V^2 + (1 - \bar{\alpha}) V^3 - V^1) + (1 - p_D) (V^2 - \underline{\alpha} V^3 - (1 - \underline{\alpha}) V^1) = \frac{(1 - \delta)s}{\delta(p_U - q_U)}, \quad (37)$$

$$p_U ((1 - \bar{\alpha}) V^2 + \bar{\alpha} V^3 - V^1) + (1 - p_U) (V^2 - (1 - \underline{\alpha}) V^3 - \underline{\alpha} V^1) = \frac{(1 - \delta)s}{\delta(p_D - q_D)}. \quad (38)$$

Writing V^k as $V^k(\omega)$, $\bar{\alpha}$ as $\bar{\alpha}(\omega)$, and $\underline{\alpha}$ as $\underline{\alpha}(\omega)$ to emphasize their dependence on ω , it follows from arguments analogous to those in the proof of Theorem 1 that $\Omega^*(3)$ is the set of parameter vectors ω given which

$$\begin{aligned} &p_D ((1 - \bar{\alpha}(\omega)) V^2(\omega) + \bar{\alpha}(\omega) V^3(\omega) - V^1(\omega)) \\ &\quad + (1 - p_D) (V^2(\omega) - (1 - \underline{\alpha}(\omega)) V^3(\omega) - \underline{\alpha}(\omega) V^1(\omega)) \geq \frac{(1 - \delta)s}{\delta(p_U - q_U)}, \\ &p_U (\bar{\alpha}(\omega) V^2(\omega) + (1 - \bar{\alpha}(\omega)) V^3(\omega) - V^1(\omega)) \\ &\quad + (1 - p_U) (V^2(\omega) - \underline{\alpha}(\omega) V^3(\omega) - (1 - \underline{\alpha}(\omega)) V^1(\omega)) \geq \frac{(1 - \delta)s}{\delta(p_D - q_D)}, \end{aligned}$$

and the boundary of $\Omega^*(3)$ contains parameter vectors given which both of these inequalities bind. This reveals that in first-best equilibria, the rewards and punishments governed by $\bar{\alpha}$ and $\underline{\alpha}$ necessarily depend on ω , ruling out uniformly first-best assignment rules.

Nonetheless, for each $\omega \in \Omega^*(3)$, we can adopt the techniques developed for our

baseline model to characterize a first-best assignment rule.

Definition 5 (First-order rotation). *For each $\omega \in \Omega^*(3)$, an assignment rule is a first-order rotation rule if it operates as follows. In each period, workers are distinctly labeled π_1, π_2 , and π_3 ; the worker labeled π_1 is assigned the desirable task and the worker labeled π_3 is assigned the undesirable task. There are four possibilities:*

1. *If both assignees produce a good output, then with probability $\bar{\alpha}(\omega)$, the worker currently labeled π_3 is next assigned label π_1 and the worker currently labeled π_2 is next labeled π_2 ; with complementary probability, the worker currently labeled π_3 is next assigned label π_2 and the worker currently labeled π_2 is next labeled π_1 . In either case, the remaining worker is next labeled π_3 .*
2. *If the worker currently assigned the undesirable task produces a good output whereas the worker currently assigned the desirable task produces a bad output, then the former worker is next labeled π_1 and the latter is next labeled π_3 . The remaining worker is next labeled π_2 .*
3. *If the worker currently assigned the undesirable task produces a bad output whereas the worker currently assigned the desirable task produces a good output, then the former worker is next labeled π_3 and the latter is next labeled π_1 . The remaining worker is next labeled π_2 .*
4. *If both assignees produce a bad output, then with probability $\underline{\alpha}(3)$, the worker currently labeled π_3 is next assigned label π_2 and the worker currently labeled π_2 is next labeled π_3 ; with complementary probability, the worker currently labeled π_3 is next assigned label π_3 and the worker currently labeled π_2 is next labeled π_2 . In either case, the remaining worker is next labeled π_1 .*

We verify that a strategy profile in which the manager plays this assignment rule and workers exert effort when assigned on path constitutes a (first-best) equilibrium. Given such strategy profile, the workers' continuation payoffs $(U(\pi_1), U(\pi_2), U(\pi_3))$

satisfy

$$\begin{aligned}
\begin{pmatrix} U(\pi_3) \\ U(\pi_1) \\ U(\pi_2) \end{pmatrix} &= (1 - \delta) \begin{pmatrix} 0 \\ x \\ r \end{pmatrix} + \delta \left[p_D p_U \begin{pmatrix} \bar{\alpha}(\omega)U(\pi_1) + (1 - \bar{\alpha}(\omega))U(\pi_2) \\ (1 - \bar{\alpha}(\omega))U(\pi_1) + \bar{\alpha}(\omega)U(\pi_2) \\ U(\pi_3) \end{pmatrix} + p_D(1 - p_U) \begin{pmatrix} U(\pi_3) \\ U(\pi_1) \\ U(\pi_2) \end{pmatrix} \right. \\
&\quad \left. + (1 - p_D)p_U \begin{pmatrix} U(\pi_1) \\ U(\pi_3) \\ U(\pi_2) \end{pmatrix} + (1 - p_U)(1 - p_D) \begin{pmatrix} \underline{\alpha}(\omega)U(\pi_2) + (1 - \underline{\alpha}(\omega))U(\pi_3) \\ (1 - \underline{\alpha}(\omega))U(\pi_2) + \underline{\alpha}(\omega)U(\pi_3) \\ U(\pi_1) \end{pmatrix} \right].
\end{aligned} \tag{39}$$

The incentive constraints are

$$\begin{aligned}
&p_D(\bar{\alpha}(\omega)U(\pi_1) + (1 - \bar{\alpha}(\omega))U(\pi_2) - U(\pi_3)) \\
&\quad + (1 - p_D)(U(\pi_1) - \underline{\alpha}(\omega)U(\pi_2) - (1 - \underline{\alpha}(\omega))U(\pi_3)) \geq \frac{(1 - \delta)s}{\delta(p_U - q_U)},
\end{aligned} \tag{40}$$

$$\begin{aligned}
&p_U((1 - \bar{\alpha}(\omega))U(\pi_1) + \bar{\alpha}(\omega)U(\pi_2) - U(\pi_3)) \\
&\quad + (1 - p_U)(U(\pi_1) - (1 - \underline{\alpha}(\omega))U(\pi_2) - \underline{\alpha}(\omega)U(\pi_3)) \geq \frac{(1 - \delta)s}{\delta(p_D - q_D)}.
\end{aligned} \tag{41}$$

By inspection of (36) and (39), $(U(\pi_1), U(\pi_2), U(\pi_3)) = (V^2(\omega), V^3(\omega), V^1(\omega))$. Because $\omega \in \Omega^*(n)$, (37) and (38) imply that (40) and (41) hold. Therefore the strategy profile constitutes a (first-best) equilibrium.

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